# Machine learning in battery management systems: a two-tier scoping review of 799 eligible studies with in-depth classification and validation audit

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**Abstract**

Machine learning (ML) has become a load-bearing component of modern battery management systems (BMS), but the validation practices behind the literature's accuracy claims are rarely examined. A structured scoping review with a two-tier evidence base is presented: a rule-based *eligible corpus* of 799 peer-reviewed journal articles (2020–2026, plus a 2019 field-defining seed), and an in-depth *classified subset* of 103 studies coded by function, method family, chemistry, data source, and validation practice, with a blind 30-paper reliability check and label-noise sensitivity ranges on the coded fields.

Union-share bibliometrics measured over the eligible corpus locate the screened literature's center of gravity in health estimation (SOH, 20% of records), state estimation (SOC, 17%), and prognostics (13%), with charging control, thermal management, cell balancing, and vehicle- or fleet-level work comparatively thin for a transportation-electrification readership. Within the classified subset, method families stratify by period — classical shallow learners, then deep sequence models, now physics-informed hybrids (the only family with a monotonically non-decreasing primary-paper trajectory) — and the first large-language-model studies appear in 2025–2026.

Three structural weaknesses recur. Only 8 of the 88 primary-application papers demonstrate their estimators online or on real vehicle or fleet data; 17 rely on within-cell or k-fold splits that leak trajectory information; and the empirical base is split between three shared public benchmarks (23 papers on Severson/Toyota, NASA, and CALCE) and private laboratory cells (36 papers) that cannot be audited at all. Physics-informed architectures, often assumed more rigorous by construction, are not demonstrably so: neither stringency contrast with data-driven peers is statistically significant, and a label-noise sensitivity analysis brackets the strong-protocol share between 40% and 59%. The review closes with a requirement-level mapping of BMS functions onto UN GTR 20, GB 38031, and ISO 26262, a positioning against prior reviews, and a twelve-item reporting checklist; the complete evidence base and analysis pipeline are deposited with a DOI.

*Keywords: machine learning; battery management systems; lithium-ion batteries; state of charge; state of health; remaining useful life; deep learning; physics-informed neural networks; transfer learning; battery safety*

## 1. Introduction

### 1.1 Electrification and the battery management problem

Lithium-ion batteries now anchor three converging transitions: the electrification of road transport, the proliferation of stationary storage in renewable-heavy grids, and the electrification of aerospace, marine, and industrial systems. In each of these settings, the battery is not a passive component but an actively managed asset whose usable

capacity, lifetime, and safety depend on operating decisions made in real time. The battery management system is the embedded cyber-physical layer that makes those decisions: it senses cell voltages, currents, and temperatures; estimates unobservable internal states; protects against over-charge, over-discharge, and thermal abuse; balances cells; and increasingly negotiates charging strategies with the surrounding power system [1].

The difficulty of the management problem stems from the fact that the quantities that matter most — state of charge (SOC), state of health (SOH), internal temperature, and the onset of degradation or fault mechanisms — are not directly measurable. They must be inferred from noisy, partially observable terminal measurements under widely varying operating conditions. Model-based approaches built on equivalent-circuit or electrochemical models have served the industry well, but they face a persistent trade-off between fidelity and computational tractability, and their parameters drift as cells age [2]. It is in this inference-and-decision layer that machine learning has made its most forceful entry.

### 1.2 The rise of machine learning in battery management

Machine learning entered battery management through estimation. Early applications treated SOC and SOH estimation as supervised regression on hand-engineered features, using support vector machines, Gaussian process regression, and shallow neural networks. Two studies then reset expectations for the whole field. Severson and colleagues showed that the entire cycle life of a lithium-ion cell could be predicted with a median test error near 9% using only features extracted from the first hundred cycles, on a public dataset of 124 cells [3]. Attia and colleagues demonstrated closed-loop Bayesian optimization of fast-charging protocols, identifying high-cycle-life protocols among 224 candidates in 16 days of closed-loop cycling rather than the more than 500 days an exhaustive search without early prediction would take [4]. Together these results reframed battery management as a data problem in which learning systems could outperform hand-tuned models both in estimation accuracy and in experiment efficiency.

The five years that followed produced a wave of published applications across every BMS function: recurrent and convolutional networks for SOC and SOH [5, 6], graph and transformer architectures for prognostics [7, 8], physics-informed neural networks that embed electrochemical and thermal priors into learning systems [9, 10], transfer learning schemes that port estimators across chemistries and vehicles [11, 12], reinforcement learning agents for charging and prognostic model selection [13], and, most recently, large language model (LLM) agents for fleet-scale risk analysis and vehicle-cloud battery management [14, 15].

### 1.3 Existing reviews and the literature gap

The review literature has grown alongside the applications, but it is fragmented along the same axes as the primary studies. An early perspective on data-driven SOC and SOH prediction [16] set the agenda; function-specific reviews cover SOC estimation [17, 18], SOH estimation [19], and prognostics [20] in depth, while method-specific reviews follow individual threads such as deep-learning state estimation [1], physics-informed learning for impedance-based diagnostics [21], or interpretable ML for prognosis [22]. Domain-anchored reviews examine fault diagnosis and safety [23, 24] and thermal management [25], alongside broader degradation-method surveys [26].

The closest in scope among recent cross-functional reviews is the performance-centric review of Prasanna et al. [27], which systematically examines 240 peer-reviewed publications spanning the whole electric-vehicle powertrain — battery management alongside charging systems, motor control, and energy-consumption prediction — and aggregates reported performance metrics (RMSE, $R^2$) into a quantitative benchmarking of algorithm families, concluding that hybrid deep-learning architectures (LSTM-DDQN, CNN-BiGRU, transformer-based frameworks) consistently achieve superior accuracy ($R^2 > 0.95$, RMSE $< 0.05$). This review differs from that work in scope, unit of analysis, and question asked. In scope, it is narrower and deeper: battery-management functions only — estimation, prognostics, charging, thermal, fault and safety, balancing — rather than the powertrain breadth, which permits function-level cuts such as the standards mapping of Section 9.4 and the per-function validation audit. In unit of analysis, it is protocol-centric rather than performance-centric: instead of pooling reported accuracy figures,

it codes how each study's validation protocol isolates (or fails to isolate) its generalization claim — cross-cell holdout versus within-cell splits, benchmark reuse versus private data, laboratory duty versus fleet duty. The two approaches are complementary, and the complementarity is pointed: an aggregation of self-reported RMSE and $R^2$ inherits, unmeasured, precisely the validation weaknesses that this review quantifies — 17 of 88 primary papers validate on information-leaking splits, and headline accuracies concentrate on three shared benchmarks — so where Prasanna et al. answer *which algorithm family performs best as reported*, this review asks *how much the reports can be trusted, and what a study must disclose for its claims to be checkable*. A third difference is evidentiary: the two-tier design here measures bibliometric shares over a released 799-record eligible corpus with record-level screening provenance, and releases the per-paper protocol audit, codebook, and re-runnable pipeline — an artifact base on which a future performance-centric synthesis could itself be audited. What remains missing from the literature even after these reviews is exactly that junction: a corpus-level account linking method family, data regime, and validation strength across management functions, which this review supplies for the BMS slice.

This gap matters because the field's self-image drives its research allocation. If, for example, most published health estimators are validated on within-cell splits of public datasets, the community's sense of progress is calibrated to a benchmark regime that differs sharply from fleet deployment. If interpretability claims are concentrated in a small method family, the safety case for certification-grade ML in production BMS is weaker than the volume of publications suggests. Answering such questions requires a corpus-level, cross-functional account of the field rather than another narrative synthesis of one function.

### 1.4 Contribution of this paper

This paper makes five contributions. First, it assembles a two-tier evidence base: a rule-based eligible corpus of 799 Crossref-verified journal articles from which field-level shares are measured within a documented capped retrieval, and an in-depth classified subset of 103 studies in which every reference's bibliographic metadata was confirmed against the Crossref registry by direct DOI resolution. Second, it develops a classification framework — BMS function × method family, extended with chemistry, data-source, and validation-practice fields — applies it uniformly to all 103 papers, and probes its reliability with a blind 30-paper re-coding under a released codebook. Third, it quantifies the corpus's composition across functions, methods, data, and validation (Section 3), anchoring function-level baselines on the quota-free eligible pool and reporting validation-practice rates with confidence intervals and label-noise sensitivity ranges. Fourth, it provides a critical, function-by-function review of what the corpus actually demonstrates (Sections 4–8), positioned explicitly against prior reviews (Section 9.7), and maps BMS functions onto UN GTR 20, GB 38031, and ISO 26262 requirements (Section 9.4). Fifth, it distills a gap-driven research agenda (Section 9) and a twelve-item reporting checklist (Section 9.9).

Fig. 1 previews the analysis, mapping the entire corpus across management functions and method families: the field's center of gravity sits squarely in the health-prognostics complex, hybrid model–data approaches now span every function, and learning-based charging control remains a concentrated Bayesian-optimization stronghold. The remainder of the paper is organized as follows. Section 2 details the review methodology: search strategy, screening procedure, inclusion and exclusion criteria, and the classification framework, together with its limitations. Section 3 presents the corpus overview and five-year bibliometric analysis. Sections 4 through 7 review machine learning for state estimation, health estimation and prognostics, charging control, and thermal management, fault detection, and safety, respectively. Section 8 examines cross-cutting directions — physics-informed learning, interpretability and uncertainty, digital twins and cloud BMS, and foundation models. Section 9 formulates research gaps and future directions, and Section 10 concludes.

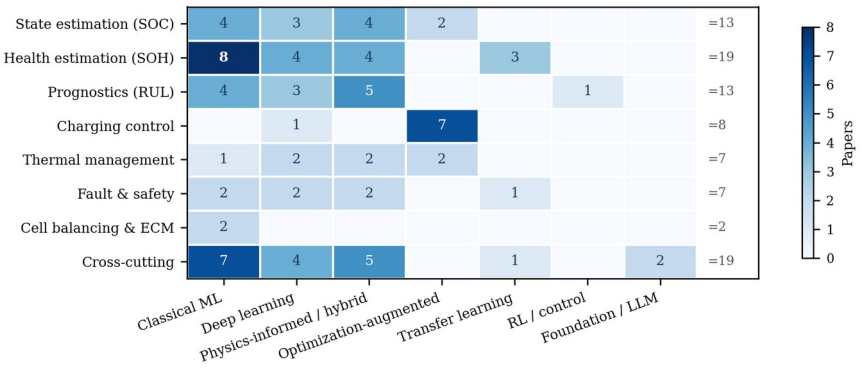



Fig. 1 The classified subset mapped across battery-management functions and method families (88 primary-application papers; see Section 2.5 for terminology). Cell values are paper counts. The 15 review-labelled papers (12 method-family reviews plus 3 review-type syntheses coded under their dominant method family, two of them physics-informed) are excluded here; the Reviews column of Table 3 states the reconciliation directly, so the physics-informed family comprises 22 primary papers plus 2 review-type syntheses (24 family-coded; Table 2 gives the primary counts).

## 2. Review Methodology

### 2.1 Scope and objective

The objective of this review is to characterize the current state and recent evolution of machine learning applied to battery management, understood as the set of estimation, protection, and control functions performed by or in support of a battery management system. The scope covers SOC estimation, SOH estimation and capacity tracking, remaining useful life (RUL) prediction and early-life prognosis, charging strategy optimization, thermal-state estimation and battery thermal management, fault diagnosis, anomaly detection and safety prediction, and cross-cutting methodological themes (physics-informed learning, transfer learning, interpretability, digital twins, cloud and fleet management, and foundation models). Battery *design* and *materials-discovery* applications of machine learning fall outside the scope except where they directly support a management function, such as manufacturing-quality prediction of cell capacity.

### 2.2 Literature search strategy

All candidate records were retrieved programmatically from the Crossref REST API between 4 and 5 September 2026 using 30 bibliographic query strings spanning the application areas listed above (for example, “state of charge estimation lithium-ion battery machine learning”, “remaining useful life prediction battery deep learning”, “physics-informed neural network lithium-ion battery”, and “large language model battery”). Queries were issued in five passes: a top-cited pass over 2020–2026 sorted by citation count; a relevance-sorted recency pass restricted to mid-2024 onward; two dedicated relevance-sorted passes restricted to 2020–2022 and to 2023; and a mid-window 2021–2023 coverage pass, which together yielded 3,000 retrieved records. After DOI-based deduplication — 1,450 duplicates removed — 1,550 unique records remained. Six landmark studies — including the early-life prediction study [3] and the prognostics review [2] — were additionally safeguarded as forced seeds: five were retrieved and verified individually because citation-sorted retrieval had crowded them out, and one was already among the top-cited hits.

Two properties of this search design should be stated plainly. First, the Crossref registry is a metadata registry, not a discovery engine of the kind Scopus or Web of Science provide; it was chosen deliberately, because the aim of this review is that every included record be individually verifiable by direct DOI resolution, which registry-native retrieval guarantees and secondary databases cannot. The cost is a recall floor: coverage depends on Crossref registration, and the usual cross-database triangulation is absent. Second, the top-cited pass inherits the biases of citation counts — it favors established, English-language, high-profile work and mechanically over-weights older publications. The three remaining passes exist to counter exactly these biases, and all trend statements in this paper

are statements about trends within this citation-weighted, English-language, journal-article sample rather than about the field in the aggregate.

### 2.3 Screening and selection procedure

The screening funnel is shown in Fig. 2. From 1,550 deduplicated records, a title-level relevance gate requiring both battery-related and machine-learning-related terminology retained 883 records and excluded 667 (predominantly adjacent-field papers — general machine-learning methods, adjacent-domain bibliometrics, and battery electrochemistry without a learning component — that surfaced through loose bibliographic matching). Venue and document-type screening then removed 84 records — conference abstracts and proceedings series (e.g., ECS Meeting Abstracts, IET/IFAC proceedings), preprint servers (SSRN), and non-scholarly venues — in line with the decision to restrict the corpus to peer-reviewed journal articles, leaving 799 eligible records. Selection then proceeds in two tiers, and the distinction carries the paper's evidentiary weight. *Tier 1 (eligible corpus).* Every one of the 799 eligible records is included, subject only to the venue screen — no function quotas, year caps, or citation thresholds are applied at this tier — and all 1,550 record-level screening decisions are released (screening_log.csv). Function tagging at Tier 1 is a rule-based keyword pass over titles with released term lists; its per-function union shares are the pool-level baselines used in Section 3.3, validated against the subset's full-text primary-function labels (71% agreement; Section 2.5). *Tier 2 (classified subset).* Because in-depth coding of chemistry, data source, and validation practice requires full-text assessment, it was applied to a 103-study subset of the eligible corpus, and the subset's construction history is disclosed in full. A stratified draw first selected 107 candidates balancing the eight management functions and publication years — within each function stratum deterministic rather than random: records ranked by Crossref citation count and taken from the top of that ranking, with the ranking walked deeper specifically to cover recent publication years that citation sorting alone would have starved. Six landmark seed studies were safeguarded at this point (one already captured by the searches, five retrieved individually), giving 112 candidates. Full-text curation then excluded 19 records — corrigenda, misindexed records, scope mismatches (for example, vehicle-level lifecycle assessment or charging-infrastructure demand modeling without a BMS function), and quota-driven trimming of over-represented function clusters — while 10 targeted additions were made from the screened pool for under-represented themes, a net change of −9 that yields the final classified corpus of 103 studies — 98 drawn from the eligible pool plus the 5 individually retrieved seeds. Per-record reasons for every exclusion and addition are carried in the released screening log and selection files, so the funnel is auditable end to end. Readers should hold the implication of the design plainly in view: the subset is curated, not sampled — function and year distributions inside the 103 partly reflect the quota and citation-ranking choices made here, not only the field's shape — so all such distributions are reported as corpus composition, the subset is never generalized to the field, and the quota-free Tier-1 measurements are stated wherever they exist (Section 3).

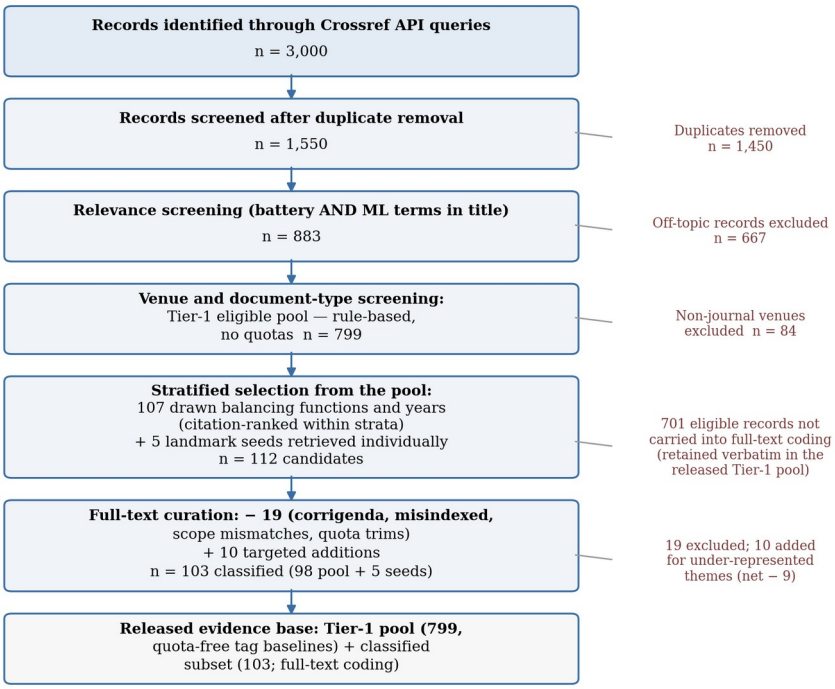



Fig. 2 Study-selection flowchart with the two-tier design. Records were retrieved from the Crossref REST API on 4–5 September 2026, deduplicated, screened at title level for battery and machine-learning relevance, then at venue and document-type level. Tier 1 includes every eligible record (799, rule-based, no quotas); Tier 2 carries the curated full-text-classified subset of 103 (98 drawn from the pool plus 5 individually retrieved seeds). Record-level provenance for all 1,550 screening decisions is released (screening_log.csv), and every included record's metadata was verified by direct DOI resolution against the Crossref registry.

### 2.4 Inclusion and exclusion criteria

Table 1 summarizes the criteria. Inclusion required: (i) a peer-reviewed journal article; (ii) a battery or battery-system subject; (iii) machine learning as a substantive component of the contribution, whether estimation, control, diagnosis, or analysis; (iv) a publication year between 2019 and 2026, with 2019 admitted only for the field-defining early-life prediction study; and (v) English language. Exclusion criteria were: review-only papers without a primary application (admitted only into the cross-cutting cluster, capped at 12 studies; three further review-type syntheses — two physics-informed, one classical — are coded under their dominant method family, so the corpus carries 15 review-labelled papers in total, per the terminology of Section 2.5), pure materials-discovery or cell-design papers, charging-infrastructure or grid-demand studies without an onboard management function, preprints and conference records, and duplicate-venue or misindexed records.

Table 1 Inclusion and exclusion criteria applied during corpus screening

| Dimension | Included | Excluded |
|---|---|---|
| Document type | Peer-reviewed journal articles | Conference abstracts/proceedings, preprints (SSRN), corrigenda |
| Subject | Batteries and battery systems with a management function | Materials discovery, cell design, grid/infrastructure demand modeling without BMS function |
| Method | ML as a substantive component (estimation, control, diagnosis, analysis) | Papers without a learning component |

| Dimension | Included | Excluded |
|---|---|---|
| Period | 2020–2026 (plus one 2019 field-defining seed) | Pre-2020 publications |
| Verification | Metadata confirmed by direct Crossref DOI resolution | Unresolvable or mismatched records |
| Language | English | Non-English |

### 2.5 Classification framework

Each corpus paper was classified along six fields: (1) *management function*, with eight values — state estimation (SOC), health estimation (SOH), prognostics (RUL), charging control, thermal management, fault and safety, cell balancing and equivalent-circuit modeling, and a cross-cutting cluster for methodological and system-level themes; (2) *method family*, with values including classical ML (Gaussian process regression, support vector machines, random forests, shallow networks), deep learning (recurrent, convolutional, attention and graph architectures), physics-informed and hybrid model–data learning, optimization-augmented learning (Bayesian optimization, surrogate-assisted and meta-heuristic schemes), reinforcement and control learning, transfer and domain adaptation, and foundation models/LLMs; (3) *chemistry* (NMC, LFP, LCO, lithium-metal, lithium–sulfur, mixed, or unspecified); (4) *primary data source* (own laboratory cycling, public datasets, laboratory impedance spectroscopy, real vehicle or fleet data, simulation, or multiple sources); (5) *validation approach* (within-cell splits, cross-cell holdout, cross-chemistry or cross-domain transfer protocols, online/real-vehicle demonstration, experimental hardware, or simulation); and (6) a short coded summary of the headline finding and the principal limitation. The validation categories of field (5) are aggregated into three stringency tiers used throughout the paper. Under the *protocol-only definition*, a *strong* protocol is one whose design isolates the generalization claim: cross-cell holdout, cross-chemistry holdout, cross-domain holdout, and online or real-vehicle demonstration. *Weak* splits leak information between train and test: within-cell splits and k-fold cross-validation without cell-level isolation. The remaining classes are *environment-only*: experimental-hardware, simulation, and single-system case-study demonstrations, which evidence an operating environment but not a data-isolation protocol; the *broader definition* of stringency additionally counts experimental-hardware demonstrations as strong. Every strong-protocol count in Sections 3 and 9 states which definition it uses, and review-labelled studies carry no primary validation. Classification was performed by the lead author with conservative coding from full texts: where a paper's specifics could not be established confidently, broader categories were assigned. Three terms are used consistently hereafter: a *corpus paper* is any of the 103 included studies; a *primary study* (or *primary-application paper*) is one of the 88 whose contribution is a primary application rather than a review; and the 15 review-labelled papers (12 method-family reviews plus 3 review-type syntheses coded under their dominant method family) are identified by the method-family and validation fields and are excluded from Fig. 1. Papers are assigned a single dominant value per field even where several apply. To probe the reliability of that coding, two co-author coders (S.S.I.A. and J.I.C., the second and third authors) later blind-re-coded a 30-paper stratified random sample (drawn by function and year band, seed-pinned) under the released codebook. The blinding was procedural rather than external: the coders are authors of this study; they coded without access to the released labels or to each other's sheets, from sheets locked and hash-tracked at completion, with a filed blinding memo documenting what each coder had opened before coding; but they are not independent of the paper, and their later author-contribution roles included verification of classifications and counts. Their access was uneven: coder B worked from 6 full texts plus 1 preprint (23 of 30 papers from abstracts only) and coder C from 2 full texts (28 from abstracts only), so the study measures abstract-level reproducibility of the coding, not full-text agreement. With those scopes stated, agreement passes the $\kappa \geq 0.66$ threshold on management function ($\kappa = 0.92$, bootstrap 95% CI 0.80–1.00), method family (0.88, CI 0.73–1.00), and primary data source (0.88, CI 0.72–1.00), and falls below it on chemistry (0.55, CI 0.16–0.83) and validation protocol (0.48, CI 0.31–0.65) — the two fields least often stated at abstract level. On the count-bearing strong-versus-non-strong bucket the two coders agree on 25 of 30 papers (raw agreement 83%, $\kappa = 0.52$, CI 0.09–0.84). Against the released labels their bucket kappas are 0.22 and 0.42, with the conservative weak-code fallback for

unreported protocols — declared in advance by both coders — driving most of that gap. No full-text dual coding was performed, so the strong/weak counts of Section 3 rest on a single full-text coding by the lead author, with two quantitative buffers: the label-noise sensitivity analysis that propagates the observed coder-versus-label disagreement into the headline counts (Section 3.4), and the complete release of per-paper labels, coder sheets, and the itemized disagreements, so that any reader can re-run, check, correct, or extend the coding — and so that a formal full-text reliability measurement can be run on the released labels.

### 2.6 Comparative synthesis approach

The synthesis proceeds on two tracks. Quantitatively, the classification fields are aggregated into cross-tabulations — function by year, method family by year, chemistry, data source, and validation practice — that characterize the corpus as a sample of the field's recent output (Section 3); every figure and every count in Section 3 was generated programmatically from the released dataset rather than compiled by hand; the full pipeline is publicly available in the repository accompanying this paper. Qualitatively, the review synthesizes each function's literature critically: what each cluster of methods assumes, what it demonstrates, and where its evidentiary support is thin (Sections 4–8). Cross-referencing the two tracks then yields the gap analysis of Section 9.

The methodology has limitations that readers should weigh. The corpus is drawn from Crossref-registered journal articles and therefore inherits that registry's coverage: influential conference papers (a significant venue for control-oriented BMS work) are excluded by design, and very recent articles registered with incomplete metadata may evade relevance screening. Title-level relevance screening can miss relevant papers whose titles avoid standard terminology, and citation-count ranking privileges English-language, high-profile venues. Classification reliability is documented on a 30-paper blind re-coding subset rather than measured corpus-wide (three of five fields above threshold, two below, with sensitivity ranges reported; the re-coding was abstract-level and co-author-coded, per Section 2.5). The classified corpus is additionally quota-curated rather than sampled: its function, year, and method distributions are conditioned on the stratified draw of Section 2.3, and the eligible pool from which it was drawn has its own retrieval-design shaping (unequal query coverage across years, Section 2.2), so neither level supports field-output trend measurement; distributional statements in Section 3 are therefore framed as corpus composition, with quota-free pool-level baselines stated where they exist. These limitations qualify the quantitative aggregates of Section 3, which should be read as a structured sample of the field's output rather than a census — and they are the reason this paper describes itself as a scoping review with bibliometric classification rather than as a meta-analysis.

## 3. Corpus Overview and Five-Year Evolution

### 3.1 Publication trend

Two layers of trend evidence are reported, and they are kept separate throughout. On the *measured* tier (eligible corpus, 799 records), the capped retrieval captures 55, 85, 139, and 223 journal articles for 2020–2023, a documented retrieval hole totaling 56 in 2024 (Section 2.2), then 126 in 2025 and — partial year, retrieval date September 2026 — 114 in 2026, plus one further early-access record date-stamped 2027 by Crossref metadata and omitted from the year chart (Fig. 3, left panel). These are counts within a capped retrieval, not census totals, and they do not support absolute-growth claims: a per-year external denominator was tested and discarded, because Crossref's reported result totals for the same queries rely on loose token-level bibliographic matches whose totals run into the hundreds of thousands and vary strongly with query phrasing, so they are not interpretable as field denominators (the test script and its output are released). What remains defensible is direction within the sample: the eligible-corpus counts rise through 2020–2023, dip at the retrieval hole, and remain high in 2025–2026.

The corpus spans 2019–2026 with the following composition: 9 papers in 2020, 13 in 2021, 13 in 2022, 22 in 2023, 20 in 2024, 18 in 2025, and 7 in the partial year 2026 (retrieval date September 2026), plus the 2019 seed (Fig. 3). This profile should be read as the output of the sampling design, not as a measurement of field output. Three design choices shape it. First, the stratified draw of Section 2.3 balanced publication years by construction, so year-to-year differences inside the subset are muted by design rather than observed. Second, the citation-ranked within-stratum

rule preferentially captures highly cited work. Third, the eligible pool itself has a retrieval-shaped year mix — the five query passes of Section 2.2 covered year windows unequally (a dedicated 2023 pass, a recency pass from mid-2024, older passes ending 2022), so the pool's own year profile (measured above) reflects query coverage as much as publication. Neither level supports a claim that field output rose, fell, or accelerated; what the composition does show, within those limits, is a broadening from single-function estimators toward cross-cutting system-level contributions in the recent quotas.

Citation weight tells a complementary story about influence. Citation counts throughout this paper are the Crossref registry's is-referenced-by-count values retrieved on 5 September 2026 and are included per paper in the released dataset; they are typically lower than Google Scholar counts for the same paper and should be compared only against themselves. The corpus accumulates roughly 18,400 recorded citations, but the distribution is steeply skewed: the median paper has 65 citations, while the 2019 seed alone holds 3,002 and the two 2020 landmark studies (Battery Lifetime Prognostics [2] and closed-loop fast charging [4]) hold 1,239 and 1,078 respectively. Median citations fall monotonically with publication year — from 441 (2020) through 141 (2022) and 46 (2024) to single digits for 2026 publications — which is partly an age effect and partly evidence that the field's attention concentrates on a small set of canonical results. For a scoping review this skew matters: narrative reviews that emphasize highly cited work systematically describe the field as it stood in 2020–2021, whereas the recent relevance-sorted portions of this corpus capture the current working frontier that citations have not yet caught up with.

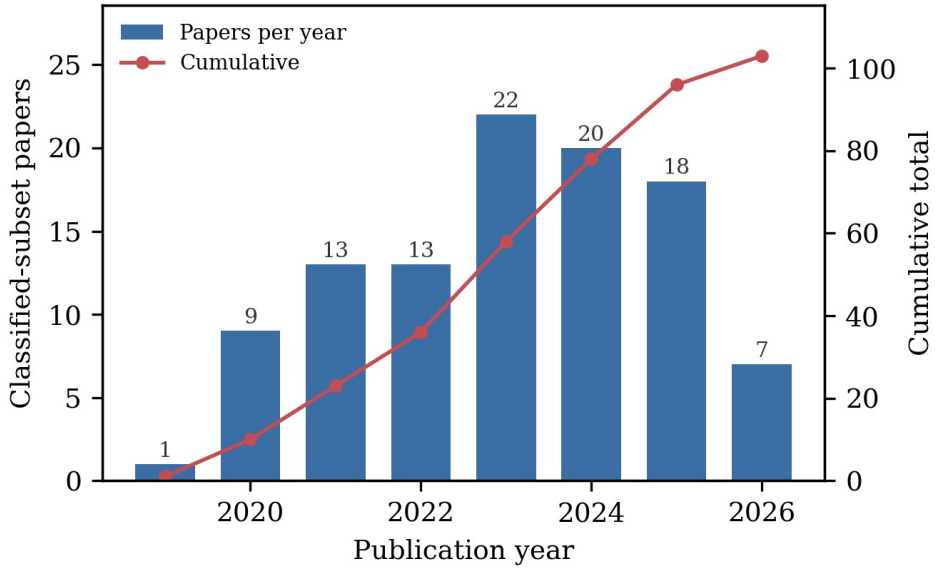



Fig. 3 Corpus composition by publication year (bars, left axis) with cumulative total (line, right axis). The 2019 seed study is included in the cumulative count. Years were balanced by the sampling design of Section 2.3, so the profile is corpus composition, not a field-output trend (Section 3.1).

### 3.2 Method-family evolution

The defining meta-finding of this review is the compositional shift in method families across the corpus (Fig. 4, Table 2) — a within-corpus observation under the quota design of Section 2.3, not a field-level growth measurement. Classical ML — Gaussian process regression, support vector machines, ensemble regressors, and shallow networks on engineered features — dominates the early corpus: 12 of the 23 papers published from 2019 through 2021. Deep sequence models (LSTM, GRU, CNN, attention and graph architectures) rise through 2022–2023 and remain the largest single family for prediction tasks thereafter. Physics-informed and hybrid model–data learning — including physics-informed neural networks (PINNs), physics-constrained autoencoders, filter-embedded networks, and multiphysics-ML fusion — emerges in 2021–2022 and grows steadily: counted year by

year, the family contributes 1, 2, 3, 3, 4, 4, and 5 primary papers in 2020–2026 — the only family with a monotonically non-decreasing primary trajectory within the subset, with 2026 a partial year at the September 2026 retrieval — so that 13 of its 22 primary papers fall in 2024–2026 and it is the largest single family among the 7 papers published in 2026 (24 family-coded papers including two review-type syntheses; Table 2 gives the primary totals; Fig. 4 the yearly counts). Optimization-augmented learning (Bayesian optimization for protocol design, surrogate-assisted multi-objective schemes, and meta-heuristic-tuned networks) peaks in 2024 (5 of its 11 papers). Transfer and domain-adaptation approaches appear consistently from 2022 and accelerate in 2024–2025 (3 of its 5 papers), reinforcement learning remains scarce but visible (1 primary paper, with RL embedded in further studies), and the first foundation-model and LLM-based studies appear in 2025–2026 (2 papers) [14, 15].

Table 2 Method families: representative techniques, strengths, and limitations. Counts are the 88 primary-application papers; the three review-type syntheses coded under a method family plus the 12 method-family reviews are accounted for in the function-by-year composition table.

| Method family | Papers | Representative techniques | Strengths | Limitations |
|---|---|---|---|---|
| Classical ML | 28 | Gaussian process regression, SVM/SVR, ensembles, shallow networks on engineered features | Calibrated uncertainty; small-data friendly; cheap to embed | Feature engineering is manual; ceilings under complex dynamics |
| Deep learning | 19 | LSTM/GRU, CNN, attention/Transformer, autoencoders, graph networks | Learns features from raw curves; strong trajectory modeling | Data-hungry; weak extrapolation; opaque |
| Physics-informed / hybrid | 22 | PINNs, physics-constrained autoencoders, filter-embedded networks, multiphysics fusion | Structure aids extrapolation; reduced data demand | Inherits simplified-model bias; harder to train |
| Optimization-augmented | 11 | Bayesian optimization, surrogate-assisted multi-objective design, meta-heuristic-tuned networks | Sample-efficient protocol discovery; explicit trade-off handling | Offline tuning cost; surrogate bias |
| Transfer / domain adaptation | 5 | Domain-adversarial training, fine-tuning across chemistry/protocol | Directly targets the generalization gap | Negative transfer; needs shared protocols |
| Reinforcement / control learning | 1 | Deep-RL prognostic model selection (also an embedded component in further studies) | Adaptive decision-making along trajectories | Training stability; scarce hardware validation |
| Foundation models / LLM | 2 | LLM incident-risk analysis; LLM-based SOH estimation with uncertainty-gated physics fallback | Reasoning over heterogeneous text/system context; transferable priors | Hallucination risk; latency and cost |
| Review / perspective | 12 | Narrative and systematic syntheses | Maps the field; identifies shared gaps | No primary validation |

Within-family dynamics deserve comment. The persistent classical-ML share is not inertia: Gaussian process regression survives because it natively returns calibrated uncertainty — the one property certification-minded applications cannot do without — and because feature-based estimators remain the cheapest to embed. The deep-learning share consolidated around recurrent and attention architectures for trajectory data and convolutional architectures for curve data, with graph networks appearing where cell-to-cell or feature-to-feature relations matter. The optimization-augmented family grew along two distinct branches: meta-heuristically tuned networks (PSO, GA, and swarm variants), which improve accuracy at a documented training cost, and Bayesian or surrogate-assisted optimization of *operating protocols*, which changes what can be experimentally discovered rather than merely what

can be predicted. Conflating these two branches, as broad-brush surveys tend to do, obscures the more consequential of the two developments.

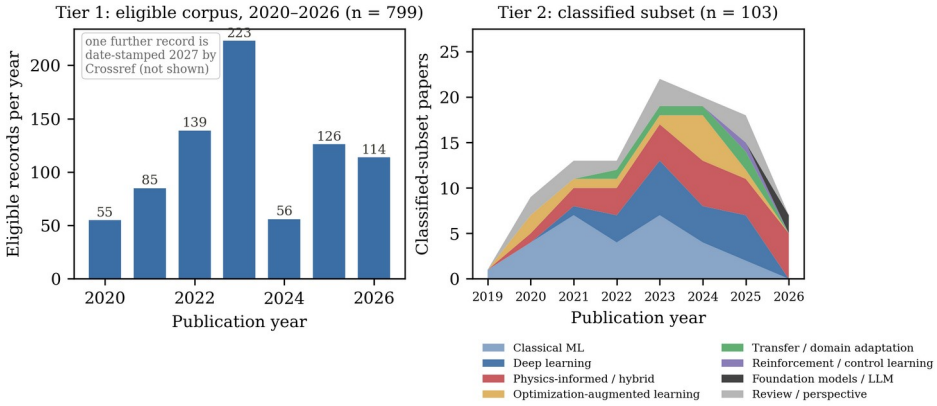



Fig. 4 Method-family composition of the classified subset by publication year (paper counts; reviews shown as their own band). Yearly totals match the corpus totals of Table 3. Composition of the curated subset under the Section 2.3 quota design; not a field-level trend measurement (Section 3.2).

Three observations qualify this trajectory. First, the families are not mutually exclusive in practice — a PSO-tuned LSTM is simultaneously deep learning and optimization-augmented learning — and the corpus classification assigns each paper to its dominant family; the composition therefore understates method mixing. Second, the rise of physics-informed learning is partly a response to the recognized brittleness of purely data-driven models under distribution shift, but physics-informed papers in the corpus are, on average, validated no more stringently than others — if anything somewhat less stringently, a point quantified in Section 3.4. Third, the foundation-model cluster is too small to represent a trend; its significance is that LLM-based management (fleet risk analysis and LLM-based health estimation, Section 8.4) is appearing in mainstream energy journals at all. A pool-level cross-check bounds these observations: among the 799 eligible records, method-theme query tags show no monotonic physics-informed trend once the retrieval windows' unequal year coverage is accounted for, so the family trajectories of this section describe the curated subset alone.

### 3.3 Application-area distribution

Functionally, the corpus is dominated by the health-prognostics complex: health estimation (20 papers) and remaining-useful-life prognostics (15) together account for 34% of the 103-paper corpus, followed by the cross-cutting cluster (24, including reviews, physics-informed methods, interpretability, transfer learning, digital twins, cloud/fleet themes, and foundation models), state estimation (17), fault and safety (9), charging control (8), thermal management (8), and cell balancing/ECM (2) (Fig. 5). Fig. 5 matches the all-corpus totals of Table 3 (103 papers); Fig. 1, by contrast, is drawn over the 88-paper primary-application corpus and excludes the 15 review-labelled papers, with the per-function reconciliation stated in that figure's caption and in Table 3's Reviews column. Because these shares are conditioned on the sampling quotas (Section 2.3), the quota-free eligible pool provides the check. Its query-theme tags — which reproduce the subset's full-text primary-function label for 53 of the 75 core-area papers found in the pool (71%, counting a hit when any of a record's tags matches; Section 2.5) — give health estimation 158, state estimation 133, prognostics 103, thermal management 99, fault and safety 96, charging control 64, and cell balancing 38 of 799 records. The pool confirms the same ordering at the top: the health-prognostics complex dominates, consistent with the commercial pull of lifetime prediction for warranties and second-life markets. It also confirms the same floor — balancing is the smallest core function at pool level as well as in the subset, so its thin representation is a property of the retrieved literature rather than an artifact of the quotas, and ML has so far contributed mainly at the level of pack-level insight rather than as a balancing controller — the corpus's most-cited balancing-related result is the demonstration that routine balancing can be dispensed with in well-managed packs, reducing hazardous lithium-plating side reactions [28]. Thermal management, by contrast, is not scarce at pool level

(99 tagged records, comparable to fault and safety): the subset's small thermal count reflects the coding scheme — thermal-runaway papers are coded under fault and safety — and the curation quotas, not field scarcity; the genuine thermal deficit is evidential rather than topical, since no subset paper yet closes the thermal control loop on hardware (Section 7.1).

Table 3 Corpus composition: management function by publication year. The Reviews column counts the 12 method-family reviews plus the 3 review-type syntheses coded under their dominant method family; Primary = Total − Reviews is the 88-paper primary-application count used in Fig. 1.

| Function | '19 | '20 | '21 | '22 | '23 | '24 | '25 | '26 | Total | Reviews | Primary |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Cell balancing & ECM | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 2 | 0 | 2 |
| Charging control | 0 | 2 | 0 | 0 | 2 | 3 | 1 | 0 | 8 | 0 | 8 |
| Cross-cutting | 0 | 1 | 0 | 6 | 8 | 3 | 2 | 4 | 24 | 5 | 19 |
| Fault & safety | 0 | 0 | 2 | 0 | 2 | 3 | 2 | 0 | 9 | 2 | 7 |
| Health estimation (SOH) | 0 | 2 | 3 | 2 | 4 | 3 | 4 | 2 | 20 | 1 | 19 |
| Prognostics (RUL) | 1 | 1 | 3 | 2 | 1 | 1 | 6 | 0 | 15 | 2 | 13 |
| State estimation (SOC) | 0 | 3 | 4 | 2 | 1 | 5 | 2 | 0 | 17 | 4 | 13 |
| Thermal management | 0 | 0 | 1 | 1 | 3 | 1 | 1 | 1 | 8 | 1 | 7 |
| Total | 1 | 9 | 13 | 13 | 22 | 20 | 18 | 7 | 103 | 15 | 88 |

Chemistry and venue concentration complete the corpus profile. NMC-type cells anchor 54 of the 103 studies and LFP 15; the remainder split across LCO, lithium-metal, lithium–sulfur, mixed panels, and unspecified chemistries — a distribution that mirrors EV market share but leaves sodium-ion, solid-state, and high-temperature systems nearly absent from the ML-management literature. Venue concentration is equally marked: the Journal of Energy Storage (15 papers), Energy (13), and Applied Energy (9) head the corpus, with Journal of Power Sources, Energy Storage Materials, and Reliability Engineering & System Safety following; IEEE venues account for 7 of the 103 papers, three of them in IEEE Transactions on Transportation Electrification. The pool shows the same shape — Journal of Energy Storage 103, Energy 61, and Batteries 45 of 799 records, with IEEE-prefixed venues at 9.0% — so for readers the practical consequence is that the ML-BMS literature's center of gravity sits in energy-systems journals rather than in machine-learning venues: partly a statement about where this work is reviewed and indexed, and partly a Crossref-registry artifact (Section 2.2), rather than evidence about how impact is achieved in this field. A final measured cut matters specifically for the transportation-electrification readership of this journal: only about 2% of eligible records (n = 17) carry fleet- or cloud-orchestration-adjacent tags, and in the classified subset only 10 of the 88 primary papers use real vehicle or fleet data as their primary source, against 36 resting on laboratory cells. The classified literature is thus predominantly cell-level and laboratory-scale — one integration level below the pack- and vehicle-level questions that UN GTR 20, GB 38031, and ISO 26262 pose; Section 9.4 maps that mismatch requirement by requirement.

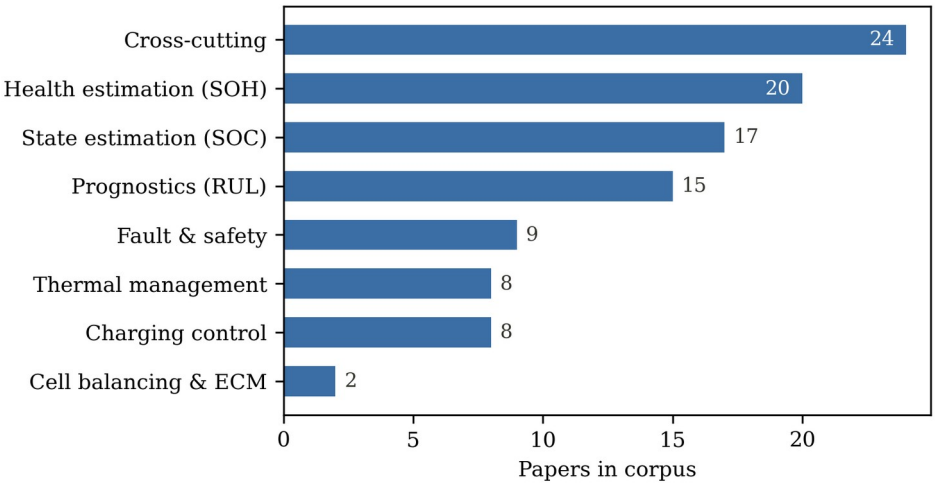



Fig. 5 Application-area distribution of the corpus. The cross-cutting cluster includes reviews, physics-informed learning, interpretability, transfer learning, digital twins, cloud and fleet management, and foundation models.

### 3.4 Data landscape and validation practice

The data underpinning the corpus are strikingly concentrated (Fig. 6, Table 4). Thirty-six papers rely primarily on laboratory cycling of the authors' own cells; the public Severson/Toyota-MIT–Stanford (MATR) cycle-life dataset alone supports 11 papers, NASA and CALCE datasets support 6 each, and laboratory impedance spectroscopy 6; 10 papers use real vehicle or fleet data as their primary data source — a distinct measure from the 8 studies whose *validation* is an online or real-vehicle demonstration; 12 are simulation-based; and 15 are reviews or perspectives that synthesize rather than generate data. Chemistry reporting is similarly skewed: 54 papers work on NMC-based cells, 15 on LFP, and the remainder on LCO, lithium-metal, lithium–sulfur, mixed panels, or unspecified chemistries.

Validation practice — arguably the field's most consequential weakness — is summarized in Fig. 6 (right panel), and it is now defined and audited explicitly (Section 2.5). Under the protocol-only definition, the *strong protocols* — cross-cell holdout (24 papers), cross-chemistry holdout (7), cross-domain holdout (3), and online or real-vehicle/fleet validation (8) — together cover 42 of the 88 primary-application papers (48%). At the weaker end, 12 papers use within-cell splits (training and testing on interleaved segments of the same cell's life), 5 use k-fold cross-validation without cross-cell isolation, and 15 are reviews without primary validation; the remaining 29 primary papers establish an operating environment (experimental hardware, 20; simulation or case-study evidence, 9) without a data-isolation protocol. The complete per-paper protocol assignment is released (protocol_audit.csv, and the valid field of papers.csv), so the 42-of-88 arithmetic can be verified row by row, and the 30-paper blind re-coding of Section 2.5 re-derived 25 of these primary-paper assignments independently. Because fleet-data *use* and online *validation* are different measures, Table 6 cross-tabulates them: of the 10 papers whose primary data are real-vehicle or fleet streams, 8 are online-validated and 2 are validated by cross-domain holdout or case-study protocols instead (a laboratory-data paper whose online claim was a desktop demonstration was recoded to experimental hardware after full-text re-check). This distribution quantifies a widely acknowledged but rarely measured problem: the majority of published accuracy claims are established under laboratory regimes that differ from deployment in chemistry coverage, temperature envelope, current-profile realism, and sensor quality — a gap that recurs throughout Sections 4–8.

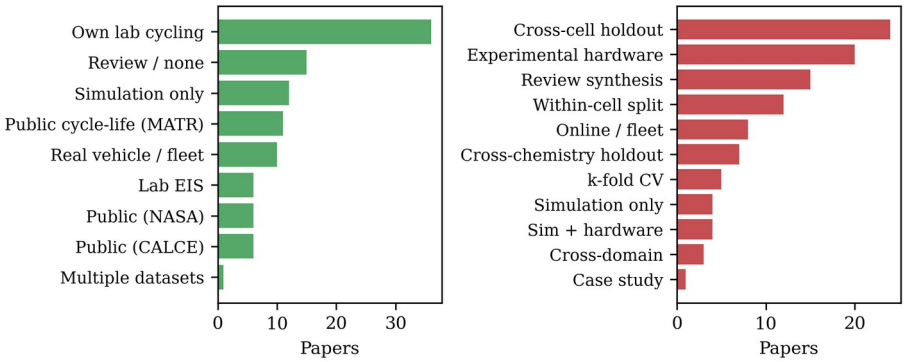



Fig. 6 Primary data sources (left) and validation approaches (right) across the corpus. Public-dataset reliance concentrates on the Severson/Toyota (MATR), NASA, and CALCE benchmarks; online or real-vehicle validation remains rare.

Table 4 Benchmark datasets and data sources recurring across the corpus

| Data source | Papers | Typical use in corpus |
|---|---|---|
| Own laboratory cycling | 36 | Primary training/testing data for estimators and protocols |
| Severson/Toyota–MIT–Stanford (MATR) | 11 | Cycle-life and early-life prediction benchmarks |
| NASA PCoE | 6 | Capacity-fade and RUL benchmarking |
| CALCE | 6 | SOH estimation and degradation modeling |
| Laboratory EIS | 6 | Impedance-based SOH/SOC and degradation-mode studies |
| Real vehicle / fleet data | 10 | Online SOH, anomaly detection, cloud BMS |
| Simulation (CFD/electrochemical) | 12 | Thermal design, charging optimization, runaway prediction |
| Review synthesis (no primary data) | 15 | Methodological and thematic reviews |
| Multiple datasets | 1 | Studies drawing on several of the above |

Validation practice deserves its own cross-tabulation because it moderates every accuracy claim in the corpus (Table 5). Cross-cell holdout — training on some cells and testing on unseen siblings — is the strongest widely used laboratory protocol (24 papers), and the transfer protocols make the strictest demands (10 papers: 7 cross-chemistry, 3 cross-domain — the seven cross-chemistry validators are the studies behind the ‘validated across chemistries’ count of the preceding paragraph). At the weaker end, within-cell splits (12 papers) and unsupervised k-fold cross-validation (5 papers) leak trajectory information between train and test sets, producing optimistic errors that do not survive cell-to-cell variability; the review literature has long warned of this, and the corpus quantifies the warning. Experimental-hardware demonstrations (20 papers) and online or real-vehicle demonstrations (8 papers) anchor the deployment end of the spectrum, and it is these — not the largest architectures — that carry the field’s deployment evidence. Crossing validation practice with method family substantiates Section 3.2’s remark with numbers: 8 of the 22 physics-informed/hybrid primary papers (36%) use a strong protocol, against 34 of the 66 other primary papers (52%) — Fisher’s exact $p = 0.32$, odds ratio 0.54 (95% CI 0.21–1.46); under the broader definition of stringency (Section 2.5) that also counts experimental-hardware demonstrations, the contrast is 13 of 22 (59%) versus 49 of 66 (74%), $p = 0.19$. Neither difference is statistically significant. To quantify how much these counts depend on the single-coder labels, a label-noise scenario analysis perturbs them directly. In each of 4,000 seeded Monte Carlo draws (seed 7), a label-flip rate is drawn from a Normal distribution centered on the observed coder-versus-label

bucket error rate of the reliability study (0.40 and 0.30 for the two coders; mean 0.35) with standard deviation 0.02, truncated at zero; the drawn rate then flips strong/non-strong bucket labels at random across the 88 primary papers, and both the overall strong share and the physics-versus-other Fisher exact test are recomputed. The strong-protocol count spans 35–52 of the 88 primary papers (39.8–59.1%), and the family contrast remains non-significant in 94.9% of draws — a physics prior does not, in itself, demonstrably buy validation rigor.

Table 5 Validation practices across the corpus

| Validation approach | Papers | Interpretation |
|---|---|---|
| Cross-cell holdout | 24 | Strongest common laboratory protocol; unseen sibling cells |
| Experimental hardware | 20 | Physical validation of estimator or controller behavior |
| Review synthesis | 15 | No primary validation applicable |
| Within-cell split | 12 | Optimistic; trajectory leakage between train and test |
| Online / real-vehicle | 8 | Deployment-relevant evidence; rarest strong protocol |
| Cross-chemistry holdout | 7 | Strictest generalization demand in corpus |
| k-fold cross-validation | 5 | Optimistic without cell-level isolation |
| Simulation (+ hardware spot-check) | 8 | Model-fidelity dependent; strongest when hardware-checked |
| Cross-domain holdout | 3 | Transfer-specific evidence |
| Case study | 1 | Single-system demonstration |

Table 6 Cross-tabulation of fleet-data use against online validation (103 classified papers). Fleet-data use describes the primary data source; online validation describes the validation protocol; their overlap is 8 of 10 fleet-data papers. A laboratory-data online demonstration was recoded to experimental hardware after full-text re-check (per the codebook absorbing rule). The right-hand column pools the environment-only validation categories defined with the classification framework.

| | Online / real-vehicle validation | All other validation protocols | Total papers |
|---|---|---|---|
| Real-vehicle / fleet data | 8 | 2 (cross-domain holdout; case study) | 10 |
| Other data sources | 0 | 93 | 93 |
| Total | 8 | 95 | 103 |

Reading Fig. 6 and Tables 4 and 5 together yields the section's headline: the field's data supply is narrow, and its validation habits are calibrated to that narrowness. Public benchmarks made cumulative progress possible — the Severson dataset alone catalyzed a family of early-prediction methods — but they now also cap the credibility of reported generalization. The studies that escape this trap are precisely those that treat data sourcing and validation design as the research contribution: the multi-cell pipeline studies, the fleet-data health estimators, and the transfer-learning frameworks.

The concentration can be stated as a metric. Across 2019–2026, 23 primary papers — 26% of the 88 primary-application papers — rest on the same three public benchmarks (MATR, NASA, CALCE; Fig. 7). These benchmarks together contain on the order of a few hundred cells, while 36 further papers rest on private laboratory data that are rarely released: the empirical base is therefore not one benchmark monoculture but two poles — a small

set of shared benchmarks onto which comparative claims crowd, and many unshared laboratory datasets over which accuracy claims cannot be audited at all. Output rose from single digits per year in 2019–2020 to 18–22 papers per year in 2023–2025, but benchmark reuse itself fluctuates between 1 and 7 papers per year with no trend, so no saturation dynamic is claimed; the structural finding is this two-pole data fragmentation, which inflates apparent comparability on the shared pole and precludes reproducibility on the private one.

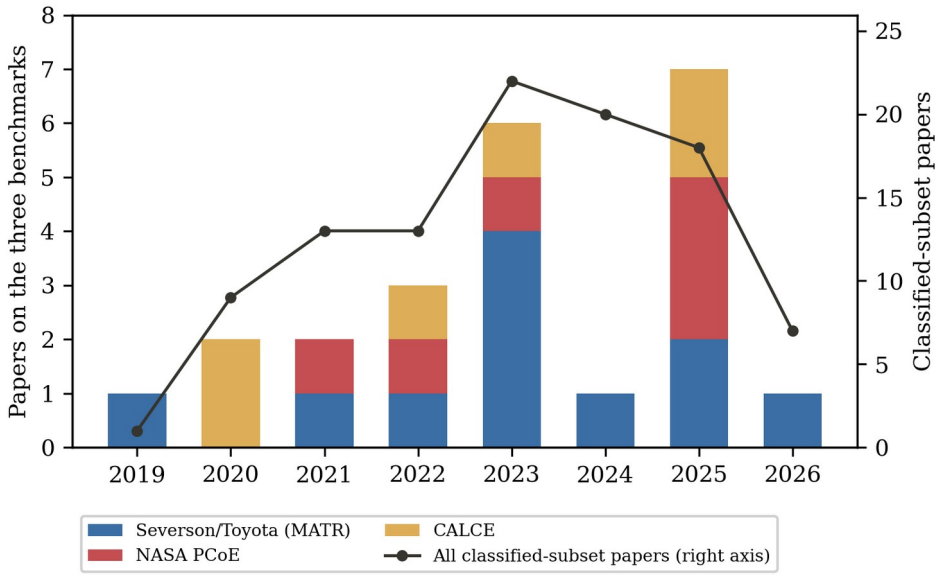



Fig. 7 Benchmark reuse versus classified-subset output. Papers in the classified subset using the three dominant public benchmarks per year (stacked bars, left axis) against classified-subset papers per year (line, right axis). Both series describe the curated 103-study subset, not the eligible corpus; benchmark reuse fluctuates year to year rather than tracking output.

### 3.5 Sensitivity analysis: does the journal-only corpus understate validation practice?

Because the corpus excludes conference proceedings — a major venue for control-oriented BMS work (CDC, ACC, IFAC, VPPC, ITEC) — the validation findings of Section 3.4 could in principle be an artifact of venue selection: if conference papers systematically demonstrate more hardware and hardware-in-the-loop validation than journals, the journal-only sample would understate the field's actual practice. To test this, we curated a sensitivity set of 14 battery papers from six control and transportation-electrification venues (ACC, ECC, CCDC, APEC, VPPC, ITEC, 2019–2025), retrieved by venue-scoped Crossref queries, restricted to proceedings articles, ranked by citation count, and verified by the same DOI-resolution procedure as the main corpus (released in full as conference_sensitivity.csv). A pre-stated inclusion rule was then applied mechanically to all retrieved proceedings papers (Table 7 lists all 14 with the rule's outcome): a paper enters the learning-path comparison only if machine learning is a substantive component of its contribution. Under this rule 8 of the 14 qualify; the remainder are conventional model-based estimators (Kalman-filter variants, multiple-model adaptive estimation, batch least squares), an EKF-critique, and one overview, and all 14 are released so that the rule can be audited rather than trusted. Validation practice was classified conservatively from the indexed abstracts into four classes: hardware/experimental stated, public dataset, simulation-level, and not stated.

Table 7 Conference-path descriptive set: all 14 battery papers retrieved from control and transportation-electrification venues (2019–2025), Crossref-verified, with validation evidence stated in their indexed abstracts. The pre-stated inclusion rule (machine learning as a substantive component of the contribution) retains 8 of the 14 for the learning-path comparison; the remainder are conventional model-based estimators or an overview and are retained here for transparency.

| Venue (year) | Function | Method (short) | Learning-based? | Validation evidence stated in indexed abstract |
|---|---|---|---|---|
| ITEC 2019 | SOC | LSTM recurrent network with transfer learning [29] | Yes | Not stated |
| VPPC 2019 | SOC | SVR + PCA with dual-polarization model [30] | Yes | Not stated |
| VPPC 2019 | SOC | Kalman filter on reduced electrochemical model [31] | No (model-based) | Simulation-level |
| ACC 2021 | SOC | Multiple-model adaptive estimation [32] | No (model-based) | Simulation-level |
| VPPC 2021 | SOC | Computationally efficient neural network [33] | Yes | Simulation-level |
| VPPC 2021 | SOH | ML ensemble [34] | Yes | Public dataset |
| ECC 2021 | SOH | Reinforcement learning + observer [35] | Yes | Not stated |
| CCDC 2022 | SOH | CNN-BiLSTM transfer learning [36] | Yes | Hardware (stated); public data |
| VPPC 2022 | Prognostics | ML on aging data [37] | Yes | Hardware (stated) |
| ITEC 2022 | SOC+SOH | Dual extended Kalman filter [38] | No (model-based) | Not stated |
| ACC 2023 | SOC+SOH | Real-time estimator + batch least squares [39] | No (model-based) | Not stated |
| APEC 2023 | Overview | Review of ML-enabled estimation [40] | Overview | Not applicable |
| ITEC 2023 | SOH | Histogram data + PCA + regression [41] | Yes | Public dataset |
| ACC 2024 | SOC+SOH | EKF critique and improvement [42] | No (model-based) | Simulation-level |

The result does not indicate a venue gap in stated validation evidence. Of the 8 learning-based conference papers, 2 (25%) state hardware or experimental validation in their indexed abstracts, 2 rest on public datasets, 1 describes simulation-level validation, and 3 state no validation basis at all — against 20 of the 88 primary-application journal papers (23%) with experimental-hardware demonstrations. Indexed abstract-level transparency about validation is, if anything, lower on the conference path. Two caveats bound this finding: conference abstracts are frequently truncated in bibliographic indexes, so the comparison measures stated evidence rather than underlying practice; and the sensitivity set is a curated sample, not a census. Within those bounds, however, the journal-only corpus does not understate the field's validation transparency — and the observation that neither venue type reports validation systematically strengthens, rather than weakens, the case for the reporting standards proposed in Section 9.9.

## 4. Machine Learning for Battery State Estimation

### 4.1 From filters to recurrent networks

SOC estimation remains the most mature BMS function, and the corpus shows how machine learning entered it. The early-corpus pattern pairs learning with classical state estimation: Gaussian process regression for pack-level SOC with native uncertainty quantification [43], and — most influentially — recurrent networks embedded in unscented Kalman filters, where the LSTM supplies a learned observation model and the filter supplies recursive correction [5]. Pure data-driven branches developed in parallel: support vector and ensemble regressors on engineered features benchmarked under automotive drive cycles [44], and simulation-informed training pipelines that substitute automotive-accurate simulated data for exhaustive cell testing [45]. Reviews of this first generation conclude that deep-learning SOC estimation is accurate on well-sampled data but suffers from weak interpretability and inconsistent benchmarking [17, 18].

A second generation tuned network structure itself. Particle-swarm-optimized LSTM improved accuracy and current-noise robustness over vanilla recurrent baselines [46], and evolutionary-assisted CNN–BiLSTM hybrids report sub-1% drive-cycle SOC errors [47]. This optimization-augmented family illustrates a recurring pattern of the corpus: gains are real but modest, and the meta-heuristic component is tuned offline while adding training cost — an investment that only pays at deployment if the estimator’s life is long. Robustness-oriented work addresses pack heterogeneity directly, jointly estimating pack SOC and cell inconsistency with balancing-aware formulations [48], large-format applications extend estimation to containerized energy-storage strings with relaxed sensing [49], unified frameworks estimate SOH and SOC jointly from partial charging data [50], and comparative benchmarks weigh accuracy against the computational budget of electrified powertrains [51, 52].

Two features of this cluster stand out against the rest of the corpus. First, it is the only function where classical estimation theory and machine learning have genuinely fused in published practice: the filter-embedded pattern [5] recurs across the five-year window in increasingly sophisticated forms, whereas health estimation and prognostics largely abandoned hybrid filtering after their Gaussian-process phase. Second, SOC papers report the corpus’s most standardized evaluation — drive-cycle profiles at room temperature with RMSE or MAE in percent SOC — which makes their benchmarks legible even where the underlying datasets remain private. The cost of that legibility is convergence: with every paper evaluating on the same standard cycles, the marginal contribution of a new architecture is shrinking, and the cluster’s research attention is shifting toward robustness conditions — noise, sensor faults, temperature, and aging shift — where the standard benchmarks are silent.

### 4.2 Hybrid and physics-informed estimation

The most substantive recent shift in SOC estimation is the move from purely learned estimators to hybrid schemes in which a filter, an equivalent-circuit model, or a physics prior constrains the learner. GA-optimized LSTM networks coupled with improved adaptive Kalman filtering reach low error under measurement noise that breaks pure-network estimators [53]; equivalent-circuit-informed ML on impedance features estimates SOC across temperatures [54]; and physics-informed recurrent structures embed battery dynamics directly into the network’s training objective [10]. These schemes trade engineering complexity for extrapolation behavior: because the physics component carries structure that does not need to be relearned, hybrid estimators degrade more gracefully outside their training envelope — the property that matters most onboard.

### 4.3 Critical assessment

The SOC literature of the corpus is accurate but inward-looking. Reported errors are consistently low on the standard drive cycles, yet 12 of the 13 primary SOC papers rest on laboratory-scale data (11 on the authors' own cells, one on laboratory impedance measurements), 9 of the 13 on a single cathode chemistry (NMC), and the thirteenth — the cluster's only operation-scale demonstration — is a simulated stationary containerized storage system rather than a vehicle [49]. The review literature itself identifies the consequence: benchmark heterogeneity makes cross-paper comparison nearly meaningless, and generalization across temperature, chemistry, and aging

state remains weakly evidenced [1]. The SOC function is, in other words, an area where the laboratory-regime estimation problem is, by reported metrics, effectively addressed — though those metrics are not comparable across papers (Section 5.6) — while the deployment problem — guaranteed accuracy on arbitrary packs over full aging — has not.

State estimation is also where the corpus's two sensing-hardware threads meet the software thread. Impedance-based SOC estimation demonstrates that electrochemical information beyond terminal voltage can drive estimation when EIS is available [54], while the ultrasonic-sensing line proposes non-electrical channels that could in principle be integrated into pack hardware at automotive cost [55]. Both remain laboratory-bound for cost and durability reasons, but they mark the boundaries of a design space — better software on existing sensors versus new sensors for better software — that the wider BMS industry is actively exploring.

## 5. Machine Learning for Health Estimation and Prognostics

### 5.1 Feature engineering and classical estimators

SOH estimation and RUL prediction form the largest functional block of the corpus (32 primary papers: 19 SOH, 13 RUL; 35 including review-labelled studies). The founding pattern is classical: extract health-sensitive features from partial charge or discharge curves, then regress with Gaussian process or similar shallow models. Incremental-capacity analysis with Gaussian process regression established the template with explicit uncertainty bounds [56]; differential thermal voltammetry extended the feature space to thermal signatures [57]; and deep Gaussian processes coupled capacity prognosis with degradation-mode identification [58]. Relaxations of the measurement burden followed: random partial charging data proved sufficient for accurate SOH [59], and charging-segment adjustment refined features for vehicle-level use [60]. The landmark systematization of this generation is the 179-cell machine-learning pipeline of Roman et al., which showed that relaxed charge-curve features support accurate, uncertainty-quantified SOH estimation across a heterogeneous commercial-cell fleet [61].

The feature taxonomy underlying this generation is worth making explicit because it explains both the cluster's strengths and its brittleness. Incremental capacity and differential voltage features transform charge curves into peaks whose positions and areas track degradation mechanisms; differential thermal voltammetry adds the temperature dimension; and partial-charging features deliberately sacrifice completeness for practicality, accepting any sufficiently informative voltage window. The approach assumes that the feature-to-health mapping is stable across the cells used for training — an assumption that holds within a cell model and laboratory but weakens across manufacturers, formats, and field profiles. The 179-cell pipeline study is important precisely because it tested that assumption at fleet scale within one commercial technology, and its success should not be over-read as license to transfer features across technologies [61].

### 5.2 Deep sequence and cross-cell models

Deep architectures entered prognostics as sequence models over capacity-fade trajectories and as representation learners over raw curves. Graph convolutional networks fuse multiple health indicators for joint SOH/RUL prediction [7]; BiLSTM–Transformer hybrids track health under the specific stress of fast charging [8]; hybrid-fusion networks target real-time prediction streams [62]; graph neural networks over partial discharge curves improve cross-cell generalization [63], and interpretable deep learning couples accuracy with transparency mechanisms [64]. A particularly important line uses real-world vehicle data rather than laboratory cycling, demonstrating that deep estimators can absorb field-data noise [6], and pack-level fusion methods now estimate SOH directly from fleet charge streams [65]. At the field's high end, a Nature Communications study showed that deep models can estimate SOH without additional degradation experiments by learning trajectory features that generalize across chemistries [66].

### 5.3 Early-life prediction

Early-life prediction — forecasting lifetime from the first cycles — is the corpus's most consequential methodological export to practice, because it compresses test times from months to days. Beyond the founding study [3], charge-process features predict cycle life across diverse duty cycles [67], ML frameworks extract lifetime information from formation-style tests [68], hybrid physics-plus-learned models give early RUL from limited cycles [69], and discharge-voltage features extend the approach to lithium-metal chemistries [70]. Board-level implementations close the loop to hardware, predicting capacity and RUL on embedded platforms in real time [71].

The economics of early-life prediction explain its rapid uptake: formation and initial cycling are already performed on every production cell, so early-prediction features are available for free at the point of manufacture, and warranty structuring, second-life triage, and cell matching all become data products. The methodological caveat is selection: cells that reach the secondary market, and cell batches that survive manufacturing screens, are conditioned samples whose early-cycle feature distributions differ from the training population. No corpus paper yet quantifies this survivorship effect, which makes it one of the more consequential open measurement problems in the cluster.

### 5.4 Physics-informed and hybrid prognostics

The prognostics cluster shows the corpus's most visible physics-informed turn. Physics-informed online degradation models support proactive health management by embedding degradation structure into learned models [72]; physics-informed hybrid reinforcement learning reduces the data demand of SOH estimation [73]; a generalizable PINN achieves cross-cell SOH estimation from limited early-cycle data [74]; EIS-informed physics-constrained networks estimate SOH from small impedance datasets [75]; and a generic physics-informed framework predicts RUL from small early-life samples across datasets [76]. Multi-feature hybrid schemes lift RUL accuracy across datasets [77], signal-decomposition hybrids sharpen capacity-fade prediction on noisy series [78], and deep reinforcement learning adaptively selects prognostic models along aging trajectories [13]. The shared motivation is clear: pure data-driven models extrapolate poorly past their training envelope, and physics priors are the cheapest available source of structure.

### 5.5 Transfer learning and cross-domain generalization

Transfer learning addresses the corpus's central weakness — generalization — directly. Domain-adversarial training adapts SOH estimators across operating conditions without target-domain labels [11]; deep transfer learning with partial charging curves carries capacity estimation across cell types [12]; transfer under fast-charging protocols extends the approach to protocol shift [79]; transfer-learning-based EIS estimation crosses chemistries [80]; and a general framework quantifies and manages the laboratory-to-fleet domain gap for trained estimators [81]. Framework-level work consolidates these threads: real-time online feature extraction from vehicle data [82] and reviews charting non-probabilistic estimation techniques [19], alongside joint SOH/RUL architectures that fuse multiple health indicators [7].

### 5.6 Critical assessment

Prognostics is where the corpus's validation problem bites hardest. The cluster's accuracy claims are established overwhelmingly on public benchmark cells: the Severson-MATR dataset alone appears in 11 corpus papers, NASA and CALCE in 12 more. Within-cell splits and k-fold protocols — which leak trajectory information between train and test — account for 10 of the 32 primary studies in the health-prognostics complex, and for 17 of the 88 primary papers in total. The transfer-learning papers are the honest exception, in that their whole object is distribution shift; but they also reveal how much shift remains unaddressed, with negative transfer cases and per-protocol fine-tuning acknowledged but rarely resolved. The mature judgment is that SOH estimation under controlled conditions is approaching incremental returns, while the prognostics problem that matters — calibrated lifetime and safety margins for arbitrary packs in the field — remains open [2, 20].

Metric heterogeneity compounds the validation problem. SOH papers report RMSE or MAE in percent capacity, sometimes normalized differently; RUL papers report cycle counts, percentages of true life, or knee-point errors defined against arbitrary end-of-life thresholds; charging papers report charge time, energy efficiency, and capacity retention after fixed cycle counts, which conflate estimation error with protocol quality. No two clusters share a metric, and within clusters the definitions vary enough that a quantitative meta-analysis of pooled accuracy — the trademark move of clinical meta-research — is not defensible for this literature. The qualitative synthesis of Sections 4–8 is therefore organized around method families, data regimes, and validation designs rather than around pooled effect sizes; this is a deliberate methodological choice, and readers should treat any cross-paper accuracy table with corresponding caution.

## 6. Machine Learning for Charging Control and Optimization

### 6.1 Closed-loop optimization of charging protocols

Charging control is the function where machine learning has changed experimental practice most directly, not merely what can be estimated. The defining result is Attia et al.: a Bayesian optimization loop over charging-protocol parameters, evaluating candidate protocols through continuous cycling, identified high-cycle-life protocols among 224 candidates in 16 days — against the more than 500 days an exhaustive search without early prediction would have taken [4]. The significance is methodological as much as practical: closed-loop learning converts protocol design from expert iteration into sample-efficient optimization. Subsequent work broadened the objective space. Ensemble surrogate models make model-based charging-strategy optimization tractable under multi-objective trade-offs [83], and data-driven Bayesian fast-charging optimization reduces charge time under explicit degradation constraints [84].

### 6.2 Learning to charge around degradation physics

A second line embeds degradation mechanisms into the charging decision itself. Machine-learning fast charging that perceives and regulates internal lithium-plating onset enables plating-free rapid protocols that violate conventional voltage-safe schedules with impunity [85]. Adaptive multistage constant-current schemes adjust stage boundaries online against measured degradation [86], and module-level optimization manages temperature gradients across packs during charging [87]. Pack-level extensions coordinate balance-aware charging for imbalanced strings [88] and multi-objective meta-heuristic schemes negotiate charge time against temperature rise and aging [89].

Scale effects are the cluster’s frontier. Cell-level protocols do not compose into pack-level schedules, because cell-to-cell variability makes the weakest cell the binding constraint; module-level optimization with temperature-gradient constraints [87] and balance-aware pack charging [88] are the corpus’s first responses. The control-theoretic gap is equally important: optimization produces open-loop schedules, while deployment wants closed-loop policies that react to measured state — which is precisely where the reinforcement-learning line, still under-represented in journal-grade hardware validation, is expected to grow.

### 6.3 Critical assessment

Charging is also where the gap between demonstration and deployment is most concrete. The corpus contains eight charging papers; of these, one is a large-scale closed-loop hardware campaign [4], three further papers validate on laboratory hardware, and the remaining four rest on simulation-level evidence (three simulation-only, one with hardware spot-checks). RL-based charging strategies — a substantial adjacent literature — are underrepresented here precisely because most are validated in simulation against electrochemical models. The unsolved problem is certifiable learning-based charging: a controller that adapts to cell-to-cell variability and aging in production vehicles must carry a safety argument that current model-free learners cannot yet make, which is why mechanism-aware schemes such as plating-perception control [85] are the most deployment-relevant results in this cluster. Table 8 collects representative studies across functions and method families.

Table 8 Representative classified-subset studies by management function and method family. Entries gloss each study as: study summary (data basis; headline contribution). References in brackets refer to the numbered list.

| Function | Method family | Representative studies (data; contribution) |
|---|---|---|
| State estimation (SOC) | Filter-embedded learning; meta-heuristic DL | LSTM+UKF embedded estimator (laboratory cells; learned observation model in a UKF) [5]; PSO-LSTM (laboratory cells; meta-heuristically tuned recurrent SOC) [46]; GA-LSTM with adaptive Kalman filtering (laboratory cells; low error under noise) [53] |
| Health estimation (SOH) | Classical ML; deep learning; transfer learning | ML pipeline over 179 cells (own cell fleet; relaxed charge-curve features) [61]; deep learning without extra degradation experiments (public MATR dataset; cross-chemistry generalization) [66]; real-world vehicle SOH (fleet data; field-noise robustness) [6] |
| Prognostics (RUL) | Classical ML; physics-informed; RL | Early-life cycle prediction (public MATR dataset; about 9% median test error) [3]; physics-informed early-cycle RUL (multiple datasets; small-data prognosis) [76]; deep-RL prognostic model fusion (public NASA dataset; adaptive model selection) [13] |
| Charging control | Bayesian optimization; mechanism-aware control | Closed-loop Bayesian protocol optimization (hardware prototype; 224 candidates in 16 days vs >500 days of exhaustive search) [4]; plating-perceiving fast charging (hardware prototype; plating-free protocols) [85]; ensemble-surrogate charging optimization (laboratory cells; multi-objective strategies) [83] |
| Thermal management | ANN surrogate; physics-informed; optimization | ANN temperature prediction (laboratory cells; low-error thermal model) [90]; PINN heat-generation estimation (simulation; embedded thermal physics) [9]; ML-assisted BTMS design (simulation; multi-objective cooling design) [91] |
| Fault & safety | Shallow ML; multiphysics + ML; transfer learning | SVM fault diagnosis (hardware prototype; sensor and connection faults) [92]; multiphysics-ML runaway prediction (simulation and hardware; module-level hazard) [93]; cross-fleet runaway-diagnosis transfer (fleet data; small-sample diagnosis) [94] |

| Function | Method family | Representative studies (data; contribution) |
|---|---|---|
| Cross-cutting | Physics-informed; XAI; digital twin; LLM | EIS diagnostics PINN review (multiple studies; paradigm synthesis) [21]; uncertainty-aware early-life prediction (public MATR dataset; calibrated forecasts) [95]; LLM-based SOH estimation with uncertainty-aware field-vehicle control (field data; subset's earliest LLM estimator; earlier LLM-based estimation exists outside the screened venues [96]) [15] |

## 7. Machine Learning for Thermal Management, Faults and Safety

### 7.1 Thermal-state estimation and thermal management

Thermal management is the corpus's smallest quantitative engineering cluster (7 primary papers; 8 including a review) but a disproportionately safety-critical one. Data-driven temperature estimation ranges from ANN models predicting cell temperature across operating conditions [90] to internal-temperature sensing under ultrahigh-rate discharge, where surface measurement fails [97]. On the design and control side, ML-surrogate models accelerate liquid-cooling plate analysis and multi-objective optimization of BTMS geometry [91, 98], experimentally validated parallel-liquid-cooling designs demonstrate the optimization loop end-to-end [99], and ANN-controlled thermal management reduces cooling parasitic load while holding temperature limits [100]. Physics-informed heat-generation estimation embeds thermal physics priors into the learning problem [9], and a state-of-the-art review consolidates the cluster [25].

The thermal cluster's papers divide cleanly by purpose, and the division clarifies where machine learning adds value. For *estimation* purposes — predicting internal or surface temperature under load — learned models replace thermocouple instrumentation that no production pack can afford, and physics-informed variants regularize the extrapolation regime where data-driven thermal models are least trustworthy [9]. For *design* purposes — sizing and arranging cooling hardware — ML surrogates amortize expensive CFD evaluation across optimization loops, turning weeks of simulation into tractable search [91, 98]. For *control* purposes — modulating cooling effort online — ANN controllers demonstrate parasitic-load reductions in simulation [100], but no subset paper yet closes this loop on hardware, leaving thermal control where charging control was five years ago.

### 7.2 Fault diagnosis and anomaly detection

Fault and safety applications (7 primary papers; 9 including reviews) divide into diagnosis of realized faults and prediction of impending hazards. On the diagnosis side, grid-search-optimized support vector machines classify sensor and connection faults [92], curvilinear-Manhattan-distance methods detect multiple pack fault types in operating fleets [101], and multi-scenario deep anomaly detection covers real EV operating envelopes [102]. Manufacturing-side inspection applies force-signal deep learning to catch lithium-plating-type defects before cells ship [103]. The hazard-prediction side is increasingly model-based: multiphysics models coupled with ML predict module-level thermal-runaway progression [93], a multiphysics-informed DeepONet predicts runaway using virtual data to augment scarce experimental events [104], and multi-source domain transfer moves thermal-runaway diagnosis across vehicle fleets with small target samples [94].

### 7.3 Critical assessment

Safety applications carry the corpus's highest stakes and its thinnest evidence base. Thermal-runaway events are rare and expensive to generate, so learning systems train on simulation or small laboratory campaigns; the DeepONet

line's virtual-data approach [104] is a rational response, but it inherits the fidelity limits of the underlying multiphysics models. The reviews in this cluster are notably candid: laboratory fault diagnosis has historically failed to transfer to real vehicles because fault signatures are masked by pack-level averaging, sensor noise, and operating diversity [23, 24]. The fleet-data anomaly-detection studies are therefore the cluster's most valuable trend, even though label scarcity for rare faults keeps evaluation largely qualitative.

## 8. Cross-Cutting Directions

### 8.1 Physics-informed learning

Physics-informed and hybrid model–data learning is the corpus's most consistently rising methodological theme (24 family-coded papers across all functions; 22 of them primary applications; the only family with a monotonically non-decreasing primary trajectory within the curated subset, Section 3.2). Beyond the prognostics applications of Section 5.4, physics-informed structures appear in state estimation [10, 105], heat-generation estimation [9], and impedance-based diagnostics, where physics-informed deep learning over electrochemical impedance spectra is consolidating into a diagnostic paradigm of its own [21]. Physics-constrained autoencoders improve health prediction over purely learned counterparts [106], and recurrent networks trained to anticipate uncertain future duty conditions address degradation prediction under distribution shift directly [107]. A parallel fusion literature combines physical and data-driven models for internal-state estimation generally [108]. The appeal is easy to state: physics priors supply exactly the extrapolation structure that pure learners lack. The corpus's caution is that most physics-informed studies embed *simplified* physics — equivalent-circuit or reduced electrochemical models — so the hybrid inherits model bias even as it gains structure.

### 8.2 Interpretability and uncertainty

A smaller but consequential cluster makes learning systems inspectable. Explainability-driven feature selection improves SOH estimator robustness via SHAP analysis [109]; interpretable ML links manufacturing coating parameters to cell capacity, connecting management analytics to production [110]; interpretable hybrid ML untangles co-evolving degradation chemistries in lithium–sulfur cells [111]; and uncertainty-aware, explainable models deliver calibrated early-life predictions [95]. Review-level retrospectives of interpretable battery prognosis now chart the theme's trajectory [22]. The state of practice remains, however, dominated by post-hoc explanation of otherwise black-box models — a necessary but not sufficient condition for the safety certification that onboard ML will eventually require.

The uncertainty thread deserves separate emphasis because it is the certification-relevant one. Gaussian-process lineages carry calibrated intervals as a first-class output [56], the uncertainty-aware early-prediction study demonstrates calibrated lifetime distributions rather than point forecasts [95], and conformal-style guarantees are beginning to appear in adjacent prognostics work. The interpretability thread, by contrast, remains dominated by post-hoc attribution (SHAP-style) analyses that explain *which features mattered* without bounding *what the model will do* on out-of-distribution inputs. For safety cases the two are not substitutes: attribution supports debugging and trust-building, while calibrated uncertainty supports decision rules — and only the latter has a route into certification argument.

### 8.3 Digital twins, cloud BMS, and new sensing

System-level architecture is emerging as a research object in its own right. Digital-twin implementations track individual battery aging through incremental learning [112] and couple LSTM surrogates with real-time thermal observation [113]; cloud-based platforms perform in-situ life prediction and classification at fleet scale from streaming charge data [114]; and second-life workflows use impedance-based accelerated SOH estimation to triage retired packs [115]. Non-electrical sensing broadens the measurement base: ultrasonic signatures processed with ML provide a SOC channel orthogonal to voltage sensing [55], impedance-spectroscopy ML supports SOC and SOH estimation despite EIS's onboard-cost problem [54, 116], and battery state-of-power estimation combines learned models with numerical search for real-time capability prediction [117].

The twin/cloud/second-life cluster carries the corpus's most concrete systems implications. Digital-twin formulations reframe estimation as continual model maintenance — the incremental-learning requirement follows from pack heterogeneity and aging drift [112] — and cloud architectures relocate heavy computation off-vehicle, at the price of connectivity dependence and data-volume engineering [114]. Second-life workflows invert the deployment direction: instead of bringing estimators to new cells, they bring cheap, accelerated assessment to aged, heterogeneous cells [115], where classification-style decisions (deploy, repurpose, recycle) tolerate coarser resolution than onboard safety estimation. These three settings differ in latency budgets, error tolerance, and update cadence — and the corpus suggests the field is beginning to design for those differences rather than for a single notion of "the BMS."

### 8.4 Foundation models and LLM-based management

The classified subset's youngest theme is generative, and its two studies divide the labor differently. Domain-specific large language models perform risk analysis of battery energy-storage incidents by synthesizing heterogeneous incident reports [14] — a reasoning-and-synthesis role. The second study, HA-LLM [15], is an LLM-based SOH estimator: it aligns resampled charging-current sequences with the LLM embedding space, injects battery degradation priors through prefix adaptation, and — most notably for safety — wraps the LLM in an uncertainty-aware module that estimates predictive uncertainty by Monte Carlo sampling and selectively substitutes a physics-consistent estimator when uncertainty is high. It is validated on one year of field data from 100 production vehicles on a vehicle-cloud platform, with an independent teardown and bench capacity calibration on an unseen vehicle model, and it remains robust in few-shot and zero-shot cross-vehicle evaluations. The published work therefore already spans both the orchestration seat and the estimator seat, and the estimator-seat study is instructive precisely because its authors treated hallucination risk as a design constraint: uncertainty-triggered fallback to a physics-consistent estimator is exactly the guardrail this review argues should be mandatory (Table 9). The appearance of both roles in mainstream energy journals in 2025–2026 marks a new phase: the management stack is beginning to absorb foundation-model components.

Because BMS software is safety-critical, this review argues for an explicit division of labor before that absorption goes further, and Table 9 makes it concrete. The table contrasts established practice for classical and deep learning estimators with what the corpus's two LLM studies actually demonstrate, and states the guardrails this review regards as mandatory before any LLM component touches a safety-relevant path. The right-hand column is normative guidance proposed here, not claims about the published studies. The notable point is that the field's first LLM estimator study [15] already implements two of those guardrails — Monte-Carlo uncertainty gating and substitution by a physics-consistent estimator — while leaving direct cell-level safety authority undefined, which is where the line should stay for now.

Table 9 Safety and verification taxonomy for foundation-model components in the BMS stack. The right-hand column is normative guidance proposed by this review, not claims about the published studies.

| Dimension | Classical / deep ML estimators (established practice) | Published LLM studies (2025–2026) | Required guardrail (this review) |
|---|---|---|---|
| Role in the stack | Calibrated numeric estimation and detection | Stationary-BESS incident risk analysis [14]; LLM-based SOH estimation with uncertainty-aware control on field vehicle data [15] | Estimation should belong to calibrated ML; the LLM SOH estimator of [15] is the earliest in the screened subset — LLM-based battery estimation exists earlier outside these venues [96] — and must show calibrated intervals before any actuation-adjacent role |

| Dimension | Classical / deep ML estimators (established practice) | Published LLM studies (2025–2026) | Required guardrail (this review) |
|---|---|---|---|
| Grounding | Trained on domain cycling data | Domain-tuned corpora and knowledge-enhanced prompting | Retrieval over verified corpora with source citations; no ungrounded recall |
| Dominant failure mode | Bounded numeric error under distribution shift | Open-ended hallucinated text | Outputs restricted to structured, checkable forms; never free text into protection logic |
| Uncertainty | Gaussian-process / conformal intervals in a subset of studies | Not reported for incident risk analysis [14]; Monte-Carlo uncertainty estimates with selective physics-estimator substitution demonstrated in the SOH estimator [15] | Calibrated intervals mandatory before any actuation-adjacent role |
| Latency and compute | Millisecond-class, onboard | Second-class, cloud-dependent | Offline analysis and planning only unless real-time evidence is shown |
| Safety authority | Bounded by physics envelopes and thresholds | Undefined in published studies | LLM components must never hold direct cell-level protection authority |

### 8.5 Manufacturing analytics and system-level synthesis

Two further cross-cutting threads connect management to the wider battery value chain. Cell-production mapping studies chart ML use across manufacturing stages from mixing to formation [118], complementing the coating-quality interpretability work [110] and manufacturing defect detection [103] — the management implication being that cell quality information is itself a BMS input. And the pack-level insight that routine cell balancing can be eliminated in well-managed packs, reducing hazardous lithium-plating side reactions, shows learned analysis reshaping a function (balancing) that classical control had considered settled [28], while ML mappings from equivalent-circuit parameters to electrochemical properties give classical BMS models a physically interpretable learned layer [119].

## 9. Research Gaps and Future Directions

Cross-referencing the bibliometric analysis of Section 3 with the function-level reviews of Sections 4–8 yields six structural gaps, discussed in Sections 9.1–9.6, summarized in Table 11, previewed against the field's milestones in Fig. 8, and then operationalized into a concrete reporting checklist in Section 9.9.

### 9.1 Validation and benchmarking standards

The corpus's validation data are unambiguous: 17 of the 88 primary papers use within-cell splits or unsupervised k-folds, and only 8 of the 88 demonstrate their systems online or on real fleets. Until cross-cell, cross-temperature, and cross-protocol holdout protocols become reporting norms, published accuracy figures will keep overstating deployable performance. The most valuable near-term contribution the community can make is standardized transfer protocols — train-on-chemistry-A/test-on-chemistry-B, train-on-lab/test-on-fleet — reported alongside headline accuracy, as already practiced by the transfer-learning and general-framework studies [80, 81].

A second standardization target is reporting itself. In our full-text reading of the corpus we did not routinely encounter disclosed split definitions, random seeds, or compute budgets; only the validation, data-source, and artifact items were systematically score-counted (Section 9.9), so this is a qualitative observation rather than a measured rate. Either way, such omissions make independent replication laborious and meta-analysis impossible

(Section 5.6). A community-endorsed reporting checklist — dataset identity and version, split protocol, uncertainty quantification, and hardware context — would cost little and would let the synthesis literature do its job.

### 9.2 Data sharing and benchmark concentration

Twenty-three primary papers rest on the same three public benchmarks (Severson-MATR 11, NASA 6, CALCE 6), and 36 use private laboratory cells whose data are not released. This two-pole fragmentation inflates apparent comparability on the shared benchmarks and hides dataset-specific failure modes, while precluding audit altogether on the private pole. Federated learning over fleet data, standardized anonymized fleet-data releases, and multi-institution cycling campaigns — in the spirit of the 179-cell pipeline [61] — would diversify the evidence base at modest incremental cost.

The economics argument strengthens the technical one. A fleet operator's marginal value from a percentage point of SOH accuracy is measured in warranty provisions and residual-value pricing, which justifies onboard compute budgets that academic prototypes routinely ignore. Quantized and pruned recurrent models, event-driven inference triggered by charge sessions rather than continuous streams, and asymmetric architectures — lightweight onboard estimators with cloud-side retraining — are the concrete design patterns emerging from the corpus's edge-aware minority [71, 82].

### 9.3 Edge deployment and onboard constraints

Only a handful of corpus works treat onboard compute, memory, or latency as first-class design constraints, notably the board-level RUL implementation [71] and real-time feature-extraction studies [82]. The gap is widening as models grow: transformer and PINN architectures trained on workstations must eventually run beside BMS microcontrollers. Quantization-aware training, neural architecture search under resource constraints, and hybrid designs that place heavy learners in the cloud and lightweight correction layers onboard are the natural research program here [15].

### 9.4 Trust, uncertainty, and safety standards

Estimates that guide protection and charging decisions need calibrated uncertainty and auditable reasoning. The corpus's uncertainty-aware [95], explainability-driven [109], and Gaussian-process lineages [56] point the way, but certification-grade ML — models whose failure modes are enumerated and bounded — remains beyond the published state of the art. Hybrid physics–data architectures are the most credible certification path because part of their behavior is analytically specified; making that argument rigorous is an open task for the PINN community.

The safety standards invoked in the introduction — UN GTR 20 (electric-vehicle safety; its Amendment 1 on thermal propagation, endorsed for WP.29 adoption in late 2026 at this review's retrieval date, offers two compliance paths: a physical test, or a risk-management approach inspired by ISO 26262 and ISO 6469-1), GB 38031 (GB 38031-2020, in force through mid-2026, and the strengthened GB 38031-2025, effective July 2026, which replaces the fire/explosion-warning criterion with no-fire/no-explosion thermal-diffusion requirements and adds bottom-impact and fast-charge-cycle tests), ISO 26262:2018 (functional safety of road-vehicle E/E systems), ISO/PAS 8800:2024 (safety and artificial intelligence — the road-vehicle extension of ISO 26262 and ISO 21448 to learned components), and ISO 21448 (SOTIF) — do not prescribe estimation algorithms, but they define what a deployed estimator or controller must be able to evidence, and Table 10 translates the subset's gaps into those terms. The mapping is this review's synthesis at requirement level rather than clause level, and it is stated so that it can be checked against the standards texts themselves: the point is not legal advice but that a validation protocol reported in a paper corresponds, or does not, to evidence a safety case can use. The inference rule behind every 'supports a standards argument' statement in this section is explicit: a paper's evidence can support such an argument only if it demonstrates performance on at least one held-out integration unit (a cross-cell holdout or stronger isolation protocol, Section 2.5) and reports the environment in which that evidence was obtained; by construction, within-cell and k-fold evidence cannot support the argument, because the safety case is about unseen cells and unseen duty. Three consequences stand out for automotive deployment. Within-cell and k-fold validation (17 primary papers)

cannot support an ISO 26262 argument, because the safety case is about unseen cells and unseen duty and the protocol never tests an unseen unit; a standards argument must therefore be *argued* from the isolation evidence, not *asserted* from headline accuracy. Thermal-propagation early-warning latency is a timing requirement, so fault-detection papers evaluated on curated laboratory windows under-report both the false-alarm rate and the detection-delay distribution that the GB 38031-2025 / GTR-20 escape window needs. And charging or thermal control strategies validated only in simulation lack the hardware-in-the-loop evidence base an automotive release process expects. The reporting checklist of Section 9.9 is deliberately aligned with this mapping — reporting an isolation argument and a validation environment is the minimum on which any standards conversation can start.

Table 10 Mapping BMS functions to requirements in UN GTR 20 (Amendment 1 thermal-propagation provisions), GB 38031-2020/2025, ISO 26262:2018 with ISO/PAS 8800:2024, and ISO 21448 (SOTIF): what deployed ML components must be able to evidence. The mapping is this review's synthesis at requirement level (not clause-level legal citation); the right-hand column gives the corresponding classified-subset primary-paper counts from Section 3.4.

| BMS function | What the standard demands of a deployed function | Current subset status |
|---|---|---|
| SOC estimation | ISO 26262:2018 treats estimation error that can enable overcharge/overdischarge as an E/E-system safety hazard to be handled through the safety lifecycle; ISO/PAS 8800:2024 extends this to learned elements (assurance argument over data, model, and known insufficiencies); UN GTR 20 / GB 38031-2025 treat overcharge as an abuse-protection requirement | 3 of 13 SOC primary papers use a strong isolation protocol; none validate on real-vehicle data; 12 rest on laboratory-scale data |
| SOH / RUL prognosis | Capacity and power fade margins underpin thermal-propagation early warning and end-of-service decisions; degradation assumptions must hold across chemistry, duty, and environment (ISO 26262 hazard analysis; ISO 21448 for performance limitations) | 20 of 32 SOH/RUL primary papers use a strong protocol; 4 use real-vehicle data; accuracy claims rest mostly on three public benchmarks (23 primary papers in the subset) |
| Fault detection | Thermal-propagation early warning must provide the escape window: GB 38031-2025 requires the warning signal but no fire/explosion beyond the triggered cell (superseding the GB 38031-2020 5-min criterion); the GTR 20 amendment proposes a ≥5-min advance warning before cabin hazards, via test or risk-management compliance path; coverage must be argued and audited (ISO/PAS 8800 assurance argument) | 2 of 7 fault/safety primary papers validate online on fleet data; detection-latency and false-alarm distributions are rarely reported |
| Charging control | Overcharge, over-temperature, and thermal-cycling limits are hard constraints under GB 38031-2020/2025; adaptive strategies must be bounded by protection envelopes and argued in the safety case | 0 of 8 charging primary papers use a strong protocol; 4 demonstrate on laboratory hardware, none online |

| BMS function | What the standard demands of a deployed function | Current subset status |
|---|---|---|
| Thermal management | Temperature-gradient and propagation-containment requirements after fast-charge or abuse (GB 38031-2025 thermal diffusion, bottom-impact, fast-charge-cycle tests); the cooling function must be verified on hardware | 0 of 7 thermal primary papers use a strong protocol; 2 exercise laboratory hardware; no learned control loop is validated on hardware |

### 9.5 Foundation-model integration

The corpus's first LLM-based management studies (2025-2026, and with LLM-based estimation prior art outside the screened venues) [14, 15] suggest a division of labor worth formalizing — with the caveat that HA-LLM [15] already occupies the estimator seat under exactly the guardrails proposed here: foundation models as reasoning, orchestration, and human-interface layers, and as estimators only where uncertainty-gated and protected by physics-fallback substitution; conventional ML as calibrated estimators; physics models as safety envelopes. Research is needed on grounding (preventing hallucinated battery knowledge), latency, and the division of safety authority between learned and non-learned components. Battery-domain foundation models pretrained on cycling corpora are a plausible medium-term development.

### 9.6 Under-served functions

Finally, the function distribution itself signals opportunity — read at both levels of the evidence base. Cell balancing is the smallest core function both in the curated subset (2 papers) and in the quota-free eligible pool (38 of 799 tagged records, Section 3.3), so its under-representation is a property of the retrieved literature rather than of the quotas; ML has so far contributed mainly at the level of pack-level insight rather than as a balancing controller [28]. Thermal management is not scarce as a theme in the pool (99 tagged records), but closed-loop thermal control has not yet been demonstrated on hardware in the subset, and the cluster remains simulation-dominated [25] — the deficit is evidential, not topical. Charging control needs certifiable adaptive protocols. These functions share a property that explains the pattern: they are *control* problems, where errors act on the battery rather than merely reporting on it, and where the evidence bar is correspondingly higher. Closing them will require the safety-case machinery of Section 9.4 as much as new algorithms.

Table 11 Structural research gaps and corresponding future directions

| Gap | Evidence in corpus | Future direction |
|---|---|---|
| Validation practice | 17 of 88 primary papers use within-cell or k-fold validation; 8 of 88 primary papers demonstrate online or fleet use | Standardized cross-cell, cross-chemistry, lab-to-fleet transfer protocols as reporting norms |
| Two-pole data fragmentation | 23 primary papers on three public benchmarks; 36 on rarely released private laboratory cells | Federated learning, anonymized fleet-data releases, multi-institution cycling campaigns |
| Onboard deployment | Few studies treat compute/memory/latency as design constraints | Quantization-aware training, resource-constrained NAS, cloud–edge hybrid estimation |
| Trust and certification | Interpretability mostly post-hoc; uncertainty rarely calibrated | Calibrated uncertainty, analytically specified hybrid architectures, safety-case methodology |
| Foundation-model integration | First two LLM-based studies appear 2025–2026 | Grounded LLM orchestration layers; domain foundation models; division of safety authority |

| Gap | Evidence in corpus | Future direction |
|---|---|---|
| Under-served functions | Balancing: smallest core function in subset (2 papers) and pool (38 of 799 tagged); closed-loop thermal control not yet shown on hardware | Certifiable adaptive control with mechanism-aware safety envelopes |

### 9.7 Position relative to prior reviews

Section 1.3 situated this review against the function-specific and method-specific reviews that dominate the field; the positioning deserves its own statement because reviewers and readers routinely ask what a new review adds when several exist. The closest in scope among recent cross-functional reviews is the performance-centric review of Prasanna et al. [27] (240 studies across the EV powertrain, aggregating reported RMSE and $R^2$); adjacent cross-functional surveys exist for degradation and estimation methods. This review is distinguished by what those studies do not attempt, and Table 12 states the comparison axis by axis: a two-tier corpus with record-level screening provenance (799 eligible records released in full, 103 classified in depth); a within-corpus measurement of validation-protocol strength across functions, with per-paper assignments released and method-family contrasts tested rather than asserted; a requirement-level mapping onto UN GTR 20, GB 38031, and ISO 26262; and a released, re-runnable analysis pipeline behind every count and figure. Where prior reviews ask *which method works best for function X*, this review asks *which claims can actually be checked — and what it would cost the reader to trust them* — which is the question a scoping review with a curated classified subset is positioned to answer, and the question that performance-centric aggregation cannot ask of itself.

Table 12 Position relative to prior reviews. “Prasanna 2026” denotes the 2026 performance-centric ML review of EV powertrains with BMS and charging scope (240 studies, aggregated reported RMSE/$R^2$). The function-specific and method-specific reviews named in Section 1.3 cover single functions or methods, so none of them addresses the axes below; the comparison is per axis, not per review. This review differs most in the validation measurement (within a curated subset), standards linkage, and released record-level artifacts.

| Axis | This review | Prior cross-functional / powertrain reviews |
|---|---|---|
| Corpus | Two-tier: 799 rule-based eligible records + 103 classified subset | Fixed study lists without record-level selection provenance |
| Validation analysis | Protocol classes defined, per-paper assignments released, method-family contrast tested with sensitivity ranges; within-corpus measurement, not a field census | Narrative remarks on data scarcity; protocol strength not quantified |
| Standards linkage | Function-to-requirement mapping vs. UN GTR 20, GB 38031-2020/2025, ISO 26262:2018, ISO/PAS 8800:2024 | Not addressed or addressed qualitatively |
| Recommendations | Twelve-item reporting checklist (BMS-ML-RC) + six research directions | General future-work statements |
| Reproducibility | Released papers.csv, screening provenance, codebook, and statistics scripts | No released record-level dataset |

### 9.8 Outlook: what the next revision cycle of this literature should produce

Concretely, and tied to the gaps above, the highest-value near-term work falls in six directions. (a) *Duty-cycle-aware validation*: public cross-cell benchmarks that pair cycling data with recorded EV drive profiles, so that cross-chemistry and cross-domain generalization can be tested against transportation duty rather than laboratory schedules. (b) *Federated fleet evidence*: privacy-preserving SOH/SOC studies over live fleets (about 2% of the eligible corpus today), with the 179-cell pipeline [61] as the template. (c) *Benchmark diversification*: multi-institution cycling campaigns two orders of magnitude larger than MATR/NASA/CALCE and released with DOI-backed artifacts, addressing the two-pole fragmentation of Section 3.4. (d) *Pack- and vehicle-level ML*: estimation validated at the integration level where UN GTR 20 and GB 38031 requirements bite, instead of cell-level evidence transplanted upward. (e) *Functionally safe learning*: ML estimators whose uncertainty handling and assurance arguments are specified against ISO 26262 safety goals and ISO/PAS 8800, with ISO 21448/SOTIF reasoning for capability limitations, making certifiable charging and thermal control reachable. And (f) *honest foundation-model integration*: retrieval-grounded LLM agents with published latency and calibration budgets, capable of the coordination role while estimation authority stays with calibrated numeric models — extending, not contradicting, the corpus's LLM estimator [15]. A guardrail on all six directions: the BMS-ML-RC checklist of Section 9.9 is designed to be attached to a paper at submission, so that the next revision cycle of this literature arrives auditable by construction.

### 9.9 A reporting checklist for machine-learning BMS studies

The most concrete artifact this review can offer is procedural. Modeled on the CONSORT and PRISMA checklists that transformed reporting in clinical and review literature, Table 13 proposes a twelve-item BMS-ML reporting checklist: the minimum metadata a study should disclose for its accuracy claims to be comparable, reproducible, and poolable in future syntheses. None of the items is onerous. Items 4, 5, and 11 operationalize the validation-protocol classes that the corpus analysis measured directly (Section 3.4); the remaining items respond to reporting omissions observed qualitatively across Sections 3–8 — undisclosed splits, uncalibrated uncertainty, unnamed datasets, unquantified compute — but only the validation, data-source, and artifact items are score-measured in this corpus, and the checklist has not been applied as an audit to all 103 studies. We suggest journals adopt it as a submission checklist and that authors attach it to their released datasets.

Table 13 Proposed BMS-ML reporting checklist (BMS-ML-RC), modeled on CONSORT/PRISMA reporting standards.

| # | Reporting item | What must be reported |
|---|---|---|
| 1 | Dataset identity | Exact name and version of every dataset (e.g., Severson-MATR, NASA, CALCE); download source |
| 2 | Cell population | Number of cells, chemistry, manufacturer, format, and batch for every training and test set |
| 3 | Sensor and sampling context | Measured channels, sampling rate, sensor class, and temperature conditions |
| 4 | Split protocol | Split type (within-cell / cross-cell / cross-chemistry / cross-domain) and isolation argument; within-cell splits must be flagged as such |
| 5 | Out-of-distribution evaluation | At least one protocol under which test conditions differ from training (chemistry, temperature, protocol, or source) |
| 6 | Baselines | At least one classical baseline alongside the proposed architecture, with identical data |
| 7 | Randomness control | Seeds or repeated-run statistics for every stochastic training result |
| 8 | Uncertainty | Calibrated uncertainty or error bars for headline claims, not point estimates alone |

| # | Reporting item | What must be reported |
|---|---|---|
| 9 | Compute budget | Training compute and inference latency/memory on the target (or a stated reference) platform |
| 10 | Failure analysis | At least one documented failure mode and its distribution (where the model is worst) |
| 11 | Validation environment | Which of the following applies: simulation, laboratory cycling, hardware prototype, online/fleet |
| 12 | Artifact availability | Data and code deposit (DOI-backed) or an explicit stated exemption |

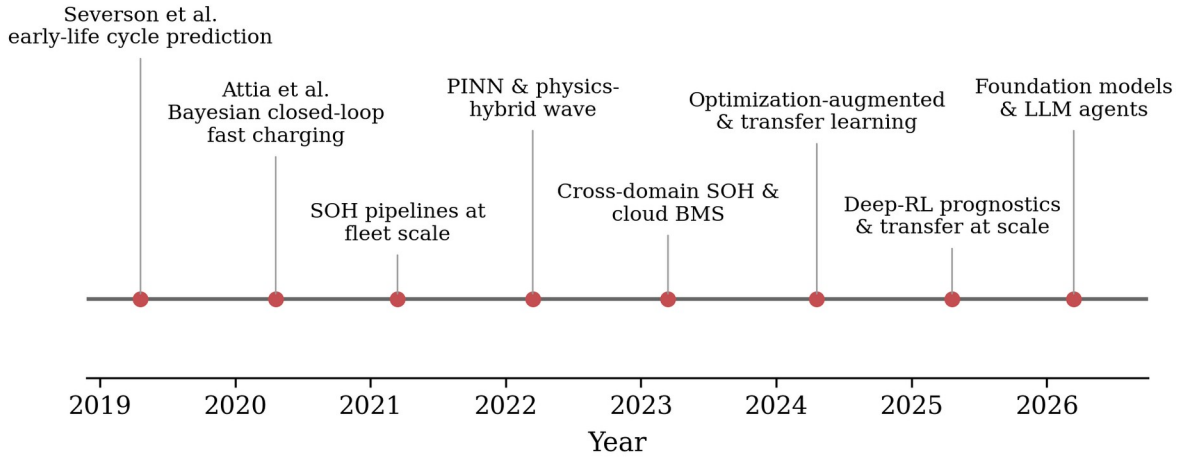



Fig. 8 Milestones of machine learning in battery management, 2019–2026, drawn from the corpus: early-life cycle prediction [3], closed-loop Bayesian fast charging [4], fleet-scale SOH pipelines [61], the physics-informed wave [9], cross-domain health estimation and cloud BMS [12, 114], optimization-augmented and transfer learning [84, 94], deep-RL prognostics [13], and the first corpus foundation-model studies [15].

## 10. Conclusion

This review has examined machine learning in battery management on a two-tier evidence base: a rule-based eligible corpus of 799 journal articles from which field-level shares are measured within a documented capped retrieval, and an in-depth classified subset of 103 studies through which method, data, and validation practice are examined with a blind 30-paper reliability check and label-noise sensitivity ranges. Three conclusions stand out. First, within the classified subset the method families stratify cleanly by period: classical feature-based learners gave way to deep sequence models, which are now themselves being displaced at the research frontier by physics-informed hybrids, optimization-augmented learning, transfer-learning regimes, and — at the very edge of the window — foundation-model components. Second, the center of gravity of application remains estimation (state, health, lifetime): measured as shares of the eligible corpus, the SOC–SOH–prognostics union covers about half of eligible records, where learning systems report laboratory-regime success — on metrics that this review argues cannot be pooled across papers (Section 5.6) — while control functions (charging, thermal, balancing) remain demonstrably transformable but under-evidenced for deployment, and vehicle- and pack-level practice stays thin relative to the level at which UN GTR 20, GB 38031, and ISO 26262 pose their requirements (Section 9.4). Third, and most critically, the proliferation of published accuracy claims has outpaced the validation evidence behind them: online or fleet-level demonstration appears in fewer than one primary study in ten, 17 of 88 primary papers use isolation-weakened splits, benchmark concentration and private-data fragmentation distort the apparent state of the

art, and the field's favored remedy — physics-informed architecture — does not demonstrably substitute for validation rigor: physics-informed papers use strong protocols no more often than purely data-driven peers (36% versus 52% under the protocol-only definition, $p = 0.32$; 59% versus 74% under the broader definition, $p = 0.19$ — not statistically significant under either, and treated here as an untested hypothesis rather than a settled result).

The near-term agenda follows directly: standardized cross-domain validation protocols; diversified and shareable fleet data; edge-deployable architectures designed under onboard constraints; calibrated uncertainty and interpretability as certification prerequisites; and a principled division of labor between foundation models, calibrated estimators, and physics-based safety envelopes. If the last five years were about discovering what machine learning can estimate about a battery, the next five will be decided by what it can be trusted to do — onboard, at fleet scale, across chemistries, and over full cell lifetimes.

**Declarations**

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**Conflicts of interest.** The authors declare no competing interests.

**Author contributions.** H.T.S. conceived the study, designed and executed the search, screening, and verification pipeline, performed the full-text classification and analysis, coded the released labels, and drafted the manuscript. S.S.I.A. and J.I.C. performed the blind 30-paper re-coding of the reliability study (coders B and C, Section 2.5) before any verification work, and contributed to corpus curation, verification of classifications and counts, and manuscript revision. All authors read and approved the final manuscript.

**Data availability.** The complete evidence base is deposited at Zenodo, DOI 10.5281/zenodo.23436202 (deposit of record; the GitHub repository of the same name is a development mirror). The deposit includes: the classified 103-study dataset (papers.csv, papers_final.json) with per-paper citation counts; the 799-record rule-based eligible pool (eligible_corpus.json, pool.csv) with per-record query-theme tags; the record-level screening log (screening_log.csv); the per-paper validation-protocol audit (protocol_audit.csv); the codebook and coding protocol (codebook.md, coding_protocol.md); the locked blind coder sheets, agreement report, itemized disagreements, and hash-tracked reliability log (corpus/reliability/); the conference sensitivity set with all 14 retrieved papers and the pre-stated inclusion rule (conference_sensitivity.csv); the withheld released labels for the reliability sample (gold_labels_30.csv); the label-noise sensitivity script and outputs (sensitivity_label_noise.py/.json); the Crossref verification audit trail (final_audit.json, external_verified.json, conference_verified.json); the seed-study register (seeds.json); and the screening statistics (stats.json).

**Code availability.** All search, screening, selection, verification, classification, reliability, sensitivity, statistics, figure-generation, and manuscript-build scripts are included in the same Zenodo deposit (DOI 10.5281/zenodo.23436202), so that every figure and every count in Sections 3 and 3.5 can be regenerated deterministically from the released data.

**Ethics approval.** Not applicable. This study did not involve human participants or animals.

**Consent to participate / for publication.** Not applicable.

**Use of artificial intelligence.** During the preparation of this manuscript the authors used the following tools, each for a stated purpose: an AI language model (GLM, Z.ai) assisted in drafting and adapting the literature-pooling, analysis, and figure-generation scripts; OpenAI ChatGPT, Grammarly, and QuillBot were used for language editing (grammar, readability, and sentence restructuring). No AI tool decided study inclusion, and no AI tool assigned or altered any classification label: all screening decisions and all released labels were made by the authors (Section 2.5 documents who coded what, and the reliability files release the coding evidence). All included references were verified against the Crossref registry by direct DOI resolution, all counts and figures were regenerated programmatically from the released dataset, and the authors reviewed and take full responsibility for the content of the published article.

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